Is there a social stigma against working wives?

Isaac Lopez-Moreno[1]

Abstract:

Goldin (1994) argued that low female labour force participation rates (FLPRs) in middle-income countries like Mexico might be related to the expansion of industrial activities, combined with a strong social stigma against wives working in blue-collar jobs. This paper offers the first empirical evaluation of three interrelated hypotheses derived from this social stigma theory using Mexican labour force surveys from 2016–2019. Although Mexico has the highest share of industrial jobs in Latin America and one of the lowest FLPRs in the region, the negative association between marriage and female employment is not confined to blue-collar jobs. Instead, Mexican wives exhibit a homogeneous lower likelihood of working across agriculture, industry, formal services and informal services. An extension of this analysis finds that Mexican women consistently report a strong negative relationship between having a partner and working across different types of industries, white-collar activities, white-collar occupations as well as high-skill and low-skill service jobs. The evidence also shows that marriage is one of the main reasons why women report having left their last job. Therefore, this paper challenges the social stigma hypothesis, since the results indicate that Mexican wives are consistently less likely to work than single women, regardless of whether it is a blue-collar or a white-collar job.

JEL Codes: J16, J12, J21

Keywords: U-shaped female labour force function, marriage and labour supply, stigma, gender division of labour.

Introduction

Growing scholarly attention is being given to the U-shaped relationship between female labour participation rates (FLPRs) and different stages of economic development across countries. Goldin (1994) suggests that a factor behind the decline in FLPRs in middle-income countries like Mexico is a strong stigma against wives working in blue-collar jobs. Specifically, the theory posits that, as the industrial sector expands, this stigma discourages married women from participating extensively in industry, as their husbands could be judged negligent if they allowed their wives to work in physically demanding jobs. Hence, countries at intermediate stages of economic development may experience a decline in FLPRs because of this social stigma.

Various studies have mentioned Goldin's social stigma theory as a potential explanation for the low FLPRs observed in cross-country studies (Burda et al., 2013; Fatima & Sultana, 2009; Gaddis & Klasen, 2013; Jayachandran, 2020; Jensen, 2012; Jütting et al., 2010; Klasen, 2019; Mammen & Paxson, 2000; Roncolato, 2016; Suleiman, 2025; Tam, 2011; Uberti & Douarin, 2023). Nevertheless, the stigma theory is more often asserted than empirically tested. To the best of my knowledge, no study has conducted a systematic empirical evaluation of its core hypotheses. Therefore, examining the social stigma theory using Mexico as a case study is relevant for several reasons. First, Mexico is a middle-income economy with low FLPRs within the U-shaped cross-country relationship (see online Appendix A, Figure 1.7). Second, Mexico has the highest share of industrial jobs in Latin America, yet one of the lowest FLPRs in the region (see online Appendix A, Figures 1.8 and 1.9). Third, there is a negative relationship between the share of industrial jobs and FLPRs across countries (see online Appendix A, Figure 1.10). In addition, recent literature suggests that the social stigma surrounding wives' participation in industrial jobs may be weaker in the twenty-first century. For instance, Olivetti (2013) finds that removing early OECD countries from her sample makes the U-shaped pattern less pronounced, and argues that married women in developing countries may be facing a lower stigma for working in blue-collar jobs, as modern manufacturing is less physically demanding. This would imply that, even if such stigma existed in the past, married women now participate in industrial employment at roughly similar rates to single women.

While the stigma theory suggests that low FLPRs in Mexico may be related to its large industrial sector and a stigma against wives working in blue-collar jobs, research on female labour participation in Mexico indicates that women's labour market participation, particularly in industry, has increased during the twenty-first century. Juhn et al. (2014) found that, after the North American Free Trade Agreement (NAFTA) came into force, Mexican women with lower educational attainment increased their participation in the export-oriented industrial sector because firms improved their manufacturing processes, adopted new technology, and reduced the physical demands of blue-collar jobs. Furthermore, Arceo-Gomez & Campos-Vazquez (2010) reported that Mexican wives are showing greater attachment to labour markets, and that their labour supply elasticities are becoming less responsive to their husbands' wages. Based on this discussion, the study poses a central question: Does Mexico's labour market exhibit evidence of a strong social stigma against wives working in blue-collar occupations?

To answer this question, the study tests three interrelated hypotheses underpinning the social stigma theory: (1) compared with single women, wives are less likely to work in agriculture and industry, although this lower likelihood does not extend to white-

[1] *PhD candidate in Development Economics at the University of Manchester (Global Development Institute).*

collar services. (2) The alleged negative relationship between being married and working in blue-collar jobs extends to male-intensive industries (mining), mixed industries (food processing), and even female-intensive industries (textiles), but it does not extend to white-collar activities. (3) The probability that married women work in industry or agriculture decreases as their husbands' educational level rises, although this pattern does not extend to white-collar jobs.

This study incorporates three important elements into the empirical evaluation of the social stigma theory. First, while Goldin discussed wives' participation in white-collar services, she overlooked the prominence of informal services in developing countries, whereas this study explicitly distinguishes between formal and informal services.[2] Second, the study does not focus solely on wives' participation across industries (mining, clothing, textiles), white-collar occupations (sales, management, clerical work), or white-collar activities (health, education, finance); it also draws on the task-content literature to assess the likelihood of wives working in service jobs demanding high-brain skills (e.g. architects), high-brawn skills (e.g. waitresses), and high-motor skills (e.g. secretaries).[3] Third, the study not only compares married and single women; it also compares different married women to assess whether their husbands' educational levels correlate with their participation in different economic sectors. Therefore, this research offers a comprehensive empirical assessment of the stigma theory.

The paper employs data from Mexico's National Survey of Occupation and Employment (ENOE) to address the research question and evaluate these three interrelated hypotheses. The ENOE is Mexico's official household survey and provides quarterly data on various labour-market indicators. The econometric model is based on a repeated cross-sectional dataset spanning four different quarters from 2016 to 2019. The regression analysis employs probit models to estimate the likelihood of married women's participation across sectors, male-intensive and female-intensive industries, white-collar activities, and both high-skill and low-skill services. Furthermore, the study uses the North American Industry Classification System (NAICS) to assess women's labour market participation by considering the characteristics of the firms where they work, and the Sistema Nacional de Clasificacion de Ocupaciones (SINCO) classification system to evaluate their labour market participation by considering the characteristics of the job that they hold.[4]

The results indicate that there is a negative relationship between marriage and female employment that is not only present in the industrial sector but also extends, with similar magnitude, to agriculture, formal services and informal services. The disaggregated findings confirm that Mexican women consistently display a strong negative association between having a partner and participating in paid work across sectors, industries, activities, occupations, and both high-skill and low-skill service jobs. The evidence further shows that marriage is among the main reasons women report leaving their most recent job. Therefore, rather than reflecting a sector-specific stigma, and irrespective of the types of jobs available in the labour market, the results suggest that the 'traditional' division of labour in married-couple households still prevails in Mexico: husbands engage in paid employment and wives are highly unlikely to work.

This finding represents a significant contribution to the literature. Researchers had assumed as a stylised fact that the negative relationship between being married and working would diminish or disappear when analysing wives' labour participation in services. However, this research shows that Mexican women consistently exhibit a negative relationship between having a partner and working across all economic sectors. Thus, the negative relationship associated with being married extends beyond a sector-specific stigma against blue-collar jobs.

The rest of the paper is organised as follows. Section 2 presents the literature review. Section 3 provides contextual background by presenting figures and tables containing relevant information on Mexico's labour market. Section 4 provides details about the database, explains the empirical strategy and the econometric model employed, and presents descriptive statistics for the selected variables. Section 5 presents the regression results, and Section 6 offers a brief discussion. Finally, Section 7 presents concluding remarks and outlines the policy implications of this study.

Literature review

Various authors have documented that a substantial portion of rising FLPRs in different countries is the result of married women entering the labour force (Blau & Kahn, 2017; Fogli & Veldkamp, 2011; Goldin, 2006). Nevertheless, Bhalotra & Fernández (2021) argue that Mexico has not followed the same pattern, as FLPRs among single women and women with a permanent partner increased in similar proportions from 1960 to 2015.[5]

---

[2] *In Mexico, a worker is considered informal if their employer does not register them with the social security system (IMSS), which is the government institution in charge of providing medical coverage, payment of wages during periods of illness, and support in cases of permanent disability resulting from work-related accidents. For the self-employed to be considered a formal worker, their business must have formal accounting practices and be registered with the tax authority and the social security system (INEGI, 2014).*

[3] *Acemoglu & Autor (2011) categorise jobs as high in cognitive tasks or high in manual tasks. Rendall (2024) classifies certain jobs as high in brain skills and brawn skills, while using motor skills to refer to jobs that require finger dexterity, manual dexterity, and motor coordination. Moreover, Autor & Dorn (2013) classify jobs using a simpler distinction between low-skill services and high-skill services.*

[4] *For more information on NACIS and SINCO, see online Appendix D.*

[5] *Bhalotra & Fernández (2021) reported that in 1960, 36% of single or divorced women in Mexico were working, compared to 71% in 2015. Meanwhile, in 1960, only 6% of women with a permanent partner were employed, whereas in 2015, it increased to 37%. In contrast, Yellen (2020) argues that in 1970, the female labour participation rate in the United States of America was 50% for single women and 40% for married women.*

A potential explanation for the limited engagement of Mexican wives in the labour market can be found in the theory developed by Goldin (1994), which posits that there is a social stigma towards wives working in industrial jobs. Since Mexico has the highest share of industrial jobs in Latin America, while also reporting one of the lowest FLPRs in the region, it is pertinent to examine this theory in the Mexican context (see online Appendix A, Figures 1.8 and 1.9). Accordingly, this literature review outlines the theoretical foundations of the industrial stigma theory and subsequently discusses studies analysing female labour participation in industrial activities in different contexts.

The stigma hypothesis

A plausible explanation for the limited engagement of Mexican wives in the labour market can be found in the theory developed by Goldin (1994). Her research shows that wives' labour force participation rates follow a U-shaped trajectory across countries, suggesting that economic development is intrinsically associated with changes in married women's labour market engagement. In low-income countries where agricultural jobs predominate, low productivity, high fertility rates and low wages result in married women participating extensively in manual labour on family farms and in in-home production. In middle-income economies where the industrial sector is expanding, wives' labour market participation declines when there is a strong social stigma against working in blue-collar occupations. Finally, in high-income countries where the service sector is expanding, the labour force participation of married women increases substantially, as there is no social stigma against wives working in white-collar jobs.

This theory places considerable emphasis on the role of social stigma against wives working in industrial jobs as a central mechanism driving the decline of FLPRs in middle-income countries. According to the theory, if a husband allows his wife to work in blue-collar jobs, society will judge him as lazy or neglectful, while having a wife working in industry also signals that their household belongs to a lower-income stratum. In general terms, Goldin (1994) posits that, as a result of this social stigma, most husbands prefer to be the sole providers for their families rather than allowing their wives to work in the industrial sector.

More specifically, Goldin proposes a series of interrelated hypotheses that, taken together, construct the basis of the social stigma theory. The first hypothesis proposes that married women are less likely to work in the industrial sector than single women. The second posits that the alleged negative relationship of wives working in industrial jobs (relative to single women) prevails not only in male-dominated sectors (such as mining and construction) but also in female-dominated industries (such as textiles and clothing manufacturing). Nonetheless, the stigma against working wives does not extend to white-collar service-sector jobs, such as clerical work, sales, education, healthcare and finance, as these are not considered physically demanding occupations. Finally, the third hypothesis argues that women married to highly educated men will show a lower propensity to take jobs in industry or agriculture than women whose husbands have lower levels of education, but this pattern does not extend to white-collar jobs.

The U-shaped female labour force function and the stigma hypothesis developed by Goldin brilliantly associate various aspects that were previously addressed separately by other researchers. Mincer (1962) and Becker (1973) discussed that, as women's educational attainment increases, the opportunity cost of time rises, which lowers the demand for children and increases female labour supply. Boserup (1970) contended that, while societies regard 'literate work' as a respectable occupation for women, they view factory work unfavourably, while also noting that jobs in industry are abhorred by most married women living in developing countries. Furthermore, Papanek (1979) and Standing (1991) argued that highly educated men can generally afford for their wives not to participate in the labour market, while the husband gains more 'family status' if his wife withdraws from employment shortly after marriage. Other researchers have also discussed the argument that wives abstaining from manual work is considered a signal of higher socioeconomic status (Munshi & Singh, 2024; Weiss & Fershtman, 1998).

These arguments align with the results presented by Hein (1986), whose research was cited by Goldin as supporting evidence for the social stigma hypothesis. Hein found that, in Mauritius, the pattern among working women was to leave their jobs in factories around the time of their marriage rather than after becoming pregnant. Additionally, the author noted that a husband's negative opinion of his wife working in a factory was a critical factor influencing married women's decisions to leave the labour market. More recently, Muñoz Boudet et al. (2013) conducted interviews about gender norms in 20 developing countries and found that deeply entrenched beliefs about which jobs are (in)appropriate for women are a typical constraint for female labour participation. Nearly 50% of the jobs mentioned in the authors' focus groups were identified as specifically for men or specifically for women, and 'feminine' roles were typically extensions of domestic tasks like cooking, teaching or nursing.

Goldin (1994) also identifies childcare as a constraint of female labour participation in industrial jobs. Her model of wives' labour supply includes childcare as one of the three uses of a woman's time. While home-based production allows women to combine work with childcare, a blue-collar job entails fixed costs, including travel expenses and the cost of finding someone to look after the children. Hence, part of the lack of women's participation in manufacturing jobs can also be explained by the fixed cost associated with entering the paid labour market outside the home.

Researchers have found that mothers with young children are more likely to seek flexible working arrangements (Berniell et al., 2021; Maloney, 2004). Moreover, Aaronson et al. (2021) concluded that agricultural work tends to offer greater flexibility because farms are often located near the home. Nayyar et al. (2021) also note that some service-sector jobs are more likely to be flexible, as new digital technologies facilitate work-from-home arrangements. Furthermore, Perry et al. (2007) discuss that informal services offer increased flexibility, control over hours and more independence. Therefore, the industrial sector may be regarded as the sector least inclined to offer flexibility to its workers, as employment must be carried out on-site and typically involves long and inflexible working hours.

In the case of Mexico, Cunningham (2001) suggests that women's labour supply patterns may be driven more by household roles than by gender. Women without partners or children exhibit labour supply patterns more comparable with those of men than with those of married women. In addition, single women without children are more likely than any other group to occupy inflexible, higher-paying positions in the formal sector. In contrast, she finds that Mexican women with children are more inclined to engage in flexible informal work to balance domestic duties with labour market participation.

Finally, Goldin also emphasises the importance of household income as a key determinant of female labour supply, while acknowledging that social stigma attached to industrial employment remains a critical barrier. When household incomes are very low, women often work on the family farm, in small-scale home production or in independent self-employment. As countries experience economic development, multiple parallel changes occur that affect wives' labour force participation. With structural transformation relocating production from household farms to agricultural factories and manufacturing firms, rising productivity generates improved wage labour prospects. Consequently, a rise in the husband's salary may reduce women's paid and unpaid labour in family enterprises through a standard income effect, especially where strong social stigmas discourage married women from engaging in manual work.

In the absence of a social stigma, women's labour supply patterns are based on a utility-maximising decision in which households compare the utility of home production and childcare against the utility of engaging in market work at the prevailing wage. In this non-stigma scenario, families evaluate the difference in utility when the wife participates in manufacturing jobs versus when she does not, without considering the loss in utility from social disgrace. By contrast, in societies where social stigma exists, households compare the utility gains from the wife's employment with the associated disutility of the stigma attached to manual or blue-collar work. If the perceived stigma exceeds the economic gains from labour market participation, the optimal choice is for wives to remain out of paid employment. Consequently, social norms against manual work magnify the negative income effect, leading to a sharper decline in women's labour force participation as household income rises.

Changing social norms and potential attenuation of social stigmas

Although the social stigma theory presents convincing arguments, social attitudes towards working wives can change over time. Therefore, theories developed in the past century may not fully apply to modern contexts. Fernández (2013) documents the cultural change in social attitudes towards the participation of married women in the labour market. She notes that, in response to the question, "Do you approve of a married woman earning money in business or industry if she has a husband capable of supporting her?", fewer than 20% of respondents agreed in 1945, compared with more than 80% in 1998.

Furthermore, Blank & Shierholz (2006) found that, in the USA, the negative effect of marital status and female labour participation in the workforce has been diminishing over time, regardless of women's skill levels. Similarly, England et al. (2004) as well as Blau & Kahn (2007) also noted that labour force participation among married women in the USA has become less influenced by their husbands' wages. Meanwhile, Arceo-Gomez & Campos-Vazquez (2010) showed that the labour supply elasticities of married women in Mexico between 1990 and 2000 exhibited a sharp decline, suggesting that their attachment to labour markets is increasing—a pattern similar to that documented in the USA.

Fernández et al. (2004) also find that wives are more likely to work if they are married to a man who was raised by a working mother during his childhood, whereas women who marry men whose mothers did not work when they were young show a lower likelihood of working. The argument is that men who had working mothers may be less averse to having a working wife, since their ideas and attitudes regarding gender roles and the gender division of work may differ from those of men whose mothers did not work. Similarly, Campos-Vazquez and Velez-Grajales (2014) estimated the probability of Mexican wives being employed based on whether their husbands had a working mother during their childhood or not. After controlling for demographic and geographic characteristics, their results showed that Mexican women whose husbands had a working mother were, on average, 15% more likely to be part of the workforce compared with those married to men who did not have a working mother.

Finally, Olivetti (2013) studied the U-shaped relationship across countries and found that excluding early developed countries from the sample made the U-shaped pattern less pronounced. To interpret this result, she argues that manufacturing may be less physically demanding in the twenty-first century and suggests that developing countries may not be stigmatising wives working in blue-collar jobs to the same extent as in early industrialised economies. Taken together, this evidence indicates that traditional social norms surrounding married women's employment may be attenuating in modern times.

Trade, technology and female labour participation in industry

In line with the previous discussion, several studies suggest that technological progress within industrial activities increases the demand for female labour, as these jobs are becoming less physically demanding and are placing greater emphasis on cognitive skills. Heath & Jayachandran (2016) argue that the shift from 'brawn-based' industries to 'light' manufacturing is one of the reasons behind the rise of female labour supply in developing countries. Weinberg (2000) also shows that the increasing use of computers accounts for over half of the growth in labour demand for female workers in the USA, since this technological change has reduced the emphasis on physical skills. This research also reveals that, although growth in female employment is positive across industries and occupations, skilled blue-collar occupations show the greatest effect. Ngai & Petrongolo (2017) also emphasise that brawn-saving technologies have reduced women's disadvantage in physically demanding tasks. Further, Rendall (2024) has shown that, as part of technological changes, the demand for labour in the industrial sector has shifted from physical labour requiring more 'brawn skills' to cognitive labour requiring more 'brain skills' in the industrial sector, and concluded that this shift has provided a more favourable environment for the integration and participation of female workers in industry.

There is also literature exploring the mechanisms of how trade liberalisation may increase female labour force participation in the industrial sector by altering firms' production technologies and demand for labour. Melitz (2003) developed a dynamic industry model showing that trade exposure leads a country's more productive firms to enter the export market, while the less productive firms continue to produce solely for the domestic market, and the least productive firms ultimately exit the market. Juhn et al. (2014) also argue that tariff reductions encourage more productive firms to enter the export market and adopt advanced technology, particularly computerised production processes that reduce the need for physically demanding skills; this results in firms replacing male blue-collar workers with female blue-collar workers.

At the country level, Ozler (2000) finds that, in the Turkish manufacturing sector during 1983–85, the share of women employed in a plant increased with the sector's export-to-output ratio. Ederington et al. (2009) also find that Colombia's trade liberalisation between 1984 and 1991 led to a significant rise in female employment, with tariff reductions driving a 6.9% increase in the proportion of female workers in export industries relative to industries where tariffs remained unchanged. Furthermore, Erten & Keski (2024) analysed the local labour demand shocks after Cambodia entered the World Trade Organization (WTO) and found that female labour participation increased in those districts facing larger tariff reductions.

In Mexico, some studies suggest that the NAFTA with the USA and Canada contributed to an increase in female labour force participation in the industrial sector. Aguayo-Tellez et al. (2010) have shown that NAFTA-driven tariff reductions expanded female-intensive sectors in Mexico, led to greater hiring of women in skilled blue-collar occupations, and contributed to a relative increase in women's wages. Juhn et al. (2014) also show that the tariff reductions experienced in 1994 following NAFTA's enactment prompted more productive firms to modernise their technology and enter the export market, leading to a shift in blue-collar employment, with many positions traditionally held by men being filled by women. Similarly, Majlesi (2016) provides evidence that positive labour demand shocks in Mexican manufacturing industries with expanded access to export markets increased women's employment opportunities. This study also links the higher female labour demand in the Mexican manufacturing sector between 2002 and 2005 to wives' greater bargaining power.

In contrast, other studies do not validate the notion that trade liberalisation consistently enhances female labour force participation. Gaddis & Pieters (2017) find no evidence that women's employment and participation expanded more than men's or that they benefited significantly from the pro-competitive effects of free trade. Furthermore, Cooray et al. (2012) examined the impact of trade on female labour force participation across a sample of 80 developing countries and found that, overall, trade has a negative effect on women's labour market participation. They suggest that this outcome may be driven by a higher incentive for women to remain out of the labour force and invest in education, as globalisation increases the skill premium. At the country level, Yu et al. (2021) find that, in China, tariff reductions have disproportionately pushed women out of the labour market relative to men, with many shifting to unpaid domestic work or early retirement, thereby reducing the female-to-male employment ratio within affected industries. Similarly, Mansour et al. (2022) analyse Peru's industrial restructuring following China's entry into the WTO, showing that rising Chinese imports led to significant employment losses among poorly educated women in Peru.

Industrial feminisation and industrial defeminisation

Having established that technological change and trade liberalisation may alter female labour participation in manufacturing, it is crucial to discuss two principal factors that characterise different patterns of women's engagement in industrial jobs. On one hand, the industrial feminisation hypothesis argues that despite the traditionally male-dominated nature of industrial employment, structural changes in the manufacturing sector have increased the demand for female workers, leading to the feminisation of the industrial workforce. Seguino & Grown (2006) argue that this trend is particularly evident in semi-industrialised economies, where women start working in low-skilled industrial jobs as firms consider that hiring women offers them a cost advantage over firms facing severe cost competition from other export-oriented economies. Other studies suggest that the growing demand for women in the industrial sector is linked to their greater agility, higher tolerance of repetitive tasks, and lower likelihood of unionising or demanding higher wages (Fernández-Kelly, 1983; Fontana, 2003; G. Standing, 1999).

In contrast, the 'industrial defeminisation hypothesis' posits that the initial stages of export-oriented industrialisation (EOI) frequently result in an increase in female employment in industry, particularly in labour-intensive manufacturing, but that, as EOI matures, the feminisation of the industrial workforce may stagnate, decline or even reverse. Caraway (2007) argues that, as export-oriented industrialisation advances, there is a process of defeminisation of the industrial workforce, as female labour shifts towards the service sector, while an increasing number of male workers re-enter the industrial sector.

Tejani & Milberg (2016) noted that, since the mid-1980s, export growth in developing countries has been associated with an industrial feminisation in some countries and an industrial defeminisation in others. Kucera & Tejani (2014) studied the patterns of feminisation and defeminisation in manufacturing activities by analysing changes in the female shares of total manufacturing employment for 36 countries at different stages of economic development from 1981 to 2008. They found that technological upgrading within labour-intensive industries (such as textiles, clothing, furniture, metal and wood) was associated with a defeminisation of the industrial workforce, as substantial advancements in technology reshaped labour composition, particularly in South Korea, Taiwan, Malaysia, Morocco and Turkey.

In line with the previous results, Seguino & Grown (2006) also argue that, in Taiwan, Hong Kong, South Korea, and Singapore, as well as in Mexico's 'maquiladoras', the share of women in manufacturing jobs has fallen in recent years. For the specific case of Mexico, Atkin (2016) studied Mexico's export-oriented industrialisation (EOI) strategy, and found that the massive expansion of export manufacturing between 1986 and 2000 generated an abundance of new low-skill formal job opportunities in the country. He also discusses the specific case of Mexico's maquiladoras, which are manufacturing firms that are legally required to export almost all their production. He noted that most of these firms were initially located in border areas close to the USA and were employing mainly women, but by 2000, one-quarter of maquiladora firms were in non-border states and half their employees were male. Taken together, the evidence indicates that industrial expansion does not produce a uniform gender pattern: while some economies exhibit rising female participation in manufacturing, others experience industrial defeminisation.

The division of labour in married-couple households.

In several married-couple households around the world, a gendered division of labour persists; husbands tend to engage primarily in paid work outside the home, while wives assume the bulk of unpaid domestic work and caregiving. My research finds no evidence that the low labour market participation of Mexican wives is primarily driven by a sector-specific stigma. Rather, it reflects a broader, persistent pattern in which married women in Mexico are highly unlikely to participate in the labour force. This subsection delves deeper into the literature discussing the pattern in which husbands assume the role of breadwinner and wives assume the role of home-maker.

Becker (1993) is a pioneer in the study of the gendered division of labour in married-couple households.He argued that family decision-making can be modelled by treating households as small 'factories' seeking to maximise a single utility function, with spouses specialising in different tasks based on their comparative advantages. He posits that such specialisation arises partly from biological differences and partly from differences in human-capital investment. As women possess a comparative advantage in childbearing, they are more likely to specialise in domestic work and assume caregiving responsibilities. Men, by contrast, have a comparative advantage in physical strength, which makes them more likely to engage in market work outside the home. In terms of human capital accumulation, Becker argues that, because women would traditionally anticipate longer absences from the labour market as a result of child-rearing, they were less likely to invest in acquiring market-oriented skills, which in turn made them more likely to become secondary earners or to specialise exclusively in unpaid household work. Consequently, biological differences also change the patterns of human capital accumulation for men and women. With these premises, the specialisation model explains the traditional pattern of gendered division of labour in married-couple households, in which women end up specialising in primary domestic activities while men specialise in market work.

While Becker's model helps to explain some mechanisms underpinning the gender division of labour in married-couple households, there are other factors that also compel women to concentrate on domestic tasks. Welter (1966) explains that one of the defining attributes of 'true womanhood'—and a central criterion by which society has judged wives—has been their adherence to domesticity, expressed through remaining within the home. Becker (2025) shows that, in contexts where men were concerned that their wives might engage in extramarital sexual relations and potentially bear children fathered by other men, societies developed social norms that stigmatised the presence of married women outside the home. Moreover, women were typically not allowed to pursue university education, and Yellen (2020) explains that limited access to education historically kept many women out of high-skilled jobs. Hence, these studies discuss how women's mobility outside the home was often stigmatised and restricted by patriarchal norms.

In the case of Mexico, Chant (1991) conducted fieldwork in three cities with contrasting labour markets—León (focused on footwear production), Puerto Vallarta (oriented to beach tourism) and Querétaro (with industrial jobs and multinational firms)—to examine changes in female labour participation shaped by differing labour market structures. Nevertheless, one of the main conclusions of her research is that, beyond the spectrum of demand variables related to varying job opportunities, household structure and household organisation are the overriding determinants of wives' participation in paid work. Some women reported direct opposition from husbands who either prohibited outside employment or viewed it as an affront to their breadwinner role. The study also indicates that, in households where women are confined to domestic tasks, men tend to exercise considerable authority over their wives' activities, thereby restricting their labour market participation.

Her research also shows that married women often cited domestic responsibilities, especially childcare and housework, as the main obstacle to seeking paid work outside the home. Meanwhile, in households where wives worked, many invited their mothers to live with them to help with housework and childcare. In other cases, sisters, cousins or nieces moved to the city to take on domestic duties in exchange for housing and the chance to continue their education. The research also documents gender expectations that cast women as self-sacrificing mothers and wives whose natural place was in the home. Many women showed signs of having internalised this ideal and believed paid work was incompatible with their roles as mothers and spouses. Hence, the research suggests that wives' labour force participation in low-income Mexican households is shaped by the gender division of household labour, the degree of control male heads of households have over their wives, the lack of childcare facilities, and deeply internalised gender norms.

Building on the previous discussion, this review emphasises that the 'traditional' division of labour in married-couple households is shaped by factors beyond biological explanations. A wide set of social and cultural norms have also historically restricted wives' participation in the labour market. Although some of these constraints have weakened over time, they are also relevant to explain the 'traditional' division of labour in many married-couple households.

Methodology

This section outlines the methodology employed to conduct an empirical assessment of the social stigma theory. It begins by detailing the database used in the study. Subsequently, it describes the empirical strategy and econometric models used to evaluate the three hypotheses related to the social stigma theory. Finally, the section concludes by presenting the descriptive statistics of the variables used in the regression analyses.

The ENOE survey is the country's largest statistical project and the principal source of statistics on Mexico's labour markets. It has been publicly released by INEGI on a quarterly basis since 2005. From 2020 onwards, a new edition of the ENOE survey was implemented in response to the COVID-19 pandemic, using the same questionnaires but allowing for both face-to-face and telephone interviews. In the fourth quarter of 2022, the emergency format was discontinued, and the regular format was reinstated in January 2023. ENOE is a panel survey with partial overlap, also known as a rotatory panel. Each survey comprises five panels, as the design follows households for five consecutive quarters. In each quarter, 20% of the sampled households have already been interviewed for five quarters, while a new set of households, also constituting 20% of the sample, replaces those that have completed the five rounds of interviews.

This study uses ENOE household surveys collected before the pandemic, as FLPRs declined sharply during the COVID-19 crisis. Survey collection during this period faced several challenges, potentially leading to biased estimates of female labour force participation. Consequently, data from 2020 and 2021 were not considered. The analysis relies on a repeated cross-sectional dataset drawn from four surveys (2016 Q1, 2017 Q2, 2018 Q3 and 2019 Q4) for two reasons. First, to include one distinct quarter from each year and account for potential seasonality. Second, to ensure that the sample consists solely of individuals interviewed once within the rotational panel. As respondents cannot be surveyed for more than five consecutive quarters, a five-quarter gap guarantees that all surveyed individuals are new respondents.

The objective of this section is to explain how the social stigma theory is assessed using data for Mexico, especially because this theory is more often asserted than empirically tested. To the best of my knowledge, Datta Gupta et al. (2020) is the only study claiming to have evaluated it. They identified Indian regions with high shares of industrial jobs and tested the hypothesis that working-age wives (25–55 years) were less likely to work when married to highly educated men working in white-collar jobs. Their results confirmed the hypothesis, and they used these findings to conclude that the stigma theory holds in India. Nevertheless, the interpretation of these results is misleading. Their analysis neither examined married women's participation across agriculture, industry, and different service activities, nor compared the probabilities of working in these sectors between married and single women. Hence, they did not effectively test the stigma theory; they simply showed that wives living in industrial regions of India and married to highly educated men working in white-collar jobs were less likely to work.

The objective of my research is to provide an in-depth empirical assessment of the social stigma theory. The following subsections outline the empirical strategy and econometric specifications used to evaluate the hypotheses underpinning this theory. The reader will observe that the next three subsections are devoted to explaining the three stages of the econometric analysis to examine different aspects of the stigma theory in a more formal and rigorous manner. Finally, the last subsection presents the descriptive statistics for all variables used across the three stages of the empirical analysis.

First stage

The first stage of the analysis evaluates one of the hypotheses underlying the social stigma theory: compared with single women, wives are less likely to work in agriculture and industry, although this lower likelihood does not extend to white-collar services. To examine this hypothesis, I estimate the probability that women are working in agriculture, industry, informal services and formal services, depending on their marital status. It is worth mentioning that, although the model below is restricted to women, equivalent regressions are estimated for men for comparative purposes. Therefore, the first hypothesis is tested using a multinomial probit model, and the econometric specification can be expressed with the following equation:

$$P(Y_i = n) = F(\beta_0 + \beta_1^n Married_i + \beta_2^n FreeUnion_i + \beta_3^n PreviouslyMarried_i + \sum_{j=6}^{j=4} \beta_j CIC_i^{(j)} + \sum_{j=10}^{j=7} \beta_j CHC_h^{(j)} + \sum_{j=13}^{j=11} \beta_j CHHC_h^{(j)} + \mu_{s,t})$$

The dependent variable 'Y' captures whether each individual *(i)* in the sample is either: 0) not working, 1) working in agriculture, 2) working in industry, 3) working in informal services,[6] 4) working in formal services, or 5) working in unspecified

[6] In Mexico, informal workers are those whose employers do not register them with the Mexican Institute of Social Security (IMSS). Distinguishing between formal and informal services is essential, as informal service jobs lack the occupational attributes, perceived status, and social legitimacy that Goldin associates with stigma-free occupations.

economic activities. The category 'not working' (Y=0) serves as the baseline for the multinomial probit estimation. Therefore, $P(Y_i = n)$ is representing the probability that an individual *(i)* is working in sector *(n)* relative to the probability of not working. In addition, 'F' denotes the multivariate normal cumulative distribution function, while 'β_0' is the intercept.

In this model, women's marital status comprises four categories: 1) married, 2) free union,[7] 3) previously married,[8] and 4) single. The model uses single women (marital status=4) as the baseline category. Thus, β_1^n, β_2^n, and β_3^n will respectively capture the relationship between working in a specific sector for women who are married, cohabiting, or previously married, relative to single women. For instance, the coefficient $\beta_1^{industry}$ will show if being married is negatively associated with the probability of working in the industrial sector, relative to single women.

'CIC' is a vector of three explanatory variables that control for the individual characteristics of each woman in the sample. 'Age' and 'age squared' are included to capture the linear and non-linear probability of working in a specific economic activity depending on the respondent's age. In addition, 'education' is a categorical variable that takes the following values: 1) primary school or less, 2) secondary school, 3) high school, 4) technical school,[9] 5) university degree.

'CHC' is a vector of four explanatory variables capturing household characteristics. The variable 'kids' indicates whether there are children under five living in the household. The variable 'urban' is a binary indicator that takes a value of 1 if the household is in an urban area and 0 if it is in a rural area. The variable 'population' is categorical and captures the population size of the household's locality. It takes the following values: 1) more than 100,000 inhabitants, 2) between 99,999 and 15,000 inhabitants, 3) between 14,999 and 2,500 inhabitants, and 4) fewer than 2,500 inhabitants. Including both variables is essential because a locality with 15,000–99,999 inhabitants may still be classified as either rural or urban, depending on INEGI's criteria. Finally, the variable 'household socioeconomic stratum' is categorised into four groups: 1) low, 2) medium-low, 3) medium-high, and 4) high. This variable was developed by INEGI using multivariate statistical techniques that incorporate 34 indicators reflecting both the human capital of household members and various household conditions, possessions, and assets. Such indicators include access to electricity and water, household overcrowding, flooring material, literacy levels of household members, access to healthcare, and ownership of items such as televisions, refrigerators and washing machines.

'CHHC' is a vector of three control variables that capture the characteristics of the household head. The vector includes the following variables: 'age of the household head', which is a continuous variable; 'education level of the household head', which is a categorical variable; and 'sex of the household head', which is a binary variable that takes the value of 1 if the household head is female, and 0 if male. To conclude, 'μ' represents the fixed effects to control for unobserved heterogeneity across time *(t)* and across states *(s)*, using Mexico City and the first quarter of 2016 as the base categories. Therefore, $t \in$ {2016 1Q, 2017 2Q, 2018 3Q, 2019 4Q} denotes the year and quarter of each survey considered in this study, while $s \in \{1, ..., 32\}$ indexes the 32 states in Mexico.

Second stage

The second stage of the econometric analysis examines another hypothesis of the social stigma theory: the alleged negative relationship between being married and working in blue-collar jobs extends to male-intensive industries (e.g. mining, iron and steel), mixed industries (food processing), and female-intensive industries (clothing and textiles), but does not extend to white-collar activities. The rationale behind Goldin's hypothesis is that the low participation of married women in the industrial sector is not simply a consequence of its male-intensive composition. Rather, her work suggests that even in female-intensive industries such as clothing or textiles, wives remain less likely to work there than single women.

In contrast, Goldin argues that this pattern does not extend to white-collar jobs. To evaluate this component of the hypothesis, the analysis examines the probability that wives are working in different types of white-collar services relative to single women. Specifically, it assesses the likelihood that wives are working in white-collar activities (health, education, finance), white-collar occupations (management, sales, clerical work), and white-collar jobs requiring different skill sets (lawyers, architects, accountants, doctors, nurses, secretaries, waitresses and hairstylists).

To assess the second hypothesis, the empirical analysis closely resembles the first stage specification, but with some key modifications. First, the empirical strategy departs from a multinomial probit model and instead estimates a set of separate probit regressions. This approach is motivated by the fact that white-collar occupations may arise in both industrial and service sectors, generating overlap across categories and violating the mutual exclusivity required in a multinomial regression. For

[7] *'Free union' refers to individuals who live permanently with their partner in the same household without being legally married.*

[8] *'Previously married' refers to women who no longer have a partner, whether as a result of widowhood, divorce, or separation.*

[9] *In Mexico, 'technical school' refers to high schools that offer technical specialisations leading to both a standard high school diploma and a professional qualification in a specific occupation. These programmes typically last four years—one more than in standard high schools—and offer basic specialisations such as electrical maintenance, carpentry, optometry, nursing, clothing manufacturing, automotive maintenance, nutrition and accounting, among others. The category also includes teacher-training schools preparing educators for rural elementary and secondary schools. As not all Mexicans have access to university education, technical schools typically provide an additional year of education and the opportunity to acquire specialised skills. Respondents who have completed any of these programmes are classified as having a technical degree.*

instance, there are managers and accountants in both mining companies and in hotels or pharmaceutical firms. Therefore, each probit model has to be estimated separately using a common specification, as shown in the following equation:

$$P(Y_{i,a}=1)=\Phi\left(\beta_0+\beta_1^a Married_i+\beta_2^a FreeUnion_i+\beta_3^a PreviouslyMarried_i+\sum_{j=6}^{j=4}\beta_j CIC_i^{(j)}+\sum_{j=10}^{j=7}\beta_j CHC_h^{(j)}+\sum_{j=13}^{j=11}\beta_j CHHC_h^{(j)}+\mu_t\right)$$

In the second stage, the dependent variables '$Y_{i,a}$' take the value of 0 if an individual *(i)* is not working and 1 if an individual *(i)* is working in a specific activity *(a)*: either an industrial subsector, a service subsector, a white-collar occupation, or a high-skill or low-skill service job. If an individual is working but not in the activity being analysed, they are excluded from the corresponding regression. This construction ensures that each probit model identifies the probability of working in a specific activity relative to the probability of not working. Therefore, $\beta_1^a Married$ will show if there is a positive or negative relationship between being married and working in a specific activity *(a)*, compared with single women.

The second modification concerns the inclusion of state fixed effects. As some states have zero observations in certain activities, state fixed effects are not considered to achieve convergence in the regression analyses, while time fixed effects are retained. Apart from these adjustments, the rest of the structure is similar to that of the first stage. Hence, in the second stage, each dependent variable is defined relative to non-participation in the labour market. They take the value of 1 if the individual is working in the specific activity *(a)* under consideration, and 0 if the individual is not working. Individuals employed in other activities are excluded from the corresponding regression. Hence, the dependent variables are defined as follows:

1. Mining – (industrial subsector): value of 1 if they work in firms dedicated to mining.
2. Iron and steel – (industrial subsector): 1 if they work in firms dedicated to iron and steel.
3. Clothing and textiles – (industrial subsector): 1 if they work in clothing and textile firms.
4. Food processing – (industrial subsector): value of 1 if they work in food processing firms.
5. Education – (service subsector): value of 1 if they work in education services.
6. Health – (service subsector): value of 1 if they work in firms that offer health services.
7. Finance – (service subsector): value of 1 if they work in firms that offer financial services.
8. Office workers – (white-collar occupation): value of 1 if they work in clerical occupations.
9. Management – (white-collar occupation): value of 1 if they work as managers.
10. Sales – (white-collar occupation): value of 1 if they work in sales.
11. Architects – (service job): value of 1 if they work as architects.
12. Lawyers – (service job): value of 1 if they work as lawyers.
13. Accounting – (service job): value of 1 if they work in accounting.
14. Doctors – (service job): value of 1 if they work as doctors.
15. Nurses – (service job): value of 1 if they work as nurses.
16. Secretaries – (service job): value of 1 if they work as secretaries.
17. Waiters – (service job): value of 1 if they work as waiters.
18. Hairstyling – (service job): value of 1 if they work as hair stylists.

Once again, all these dependent variables are equal to 0 if an individual is not working and 1 if the individual is working in that specific activity *(a)*. Therefore, $P(Y_{i,a}=1)$ represents the probability that an individual *(i)* is working in one of these activities *(a)*, relative to the probability of not working. In this model, $\beta_1^a Married$ will show if there is a positive or negative relationship between being married and working in a specific activity *(a)*, compared with single women. This approach enables the assessment of the second hypothesis: the alleged negative relationship between being married and working in blue-collar jobs extends to different industrial activities (female-intensive or male-intensive) but does not extend to white-collar services.

The construction of each independent variable was relatively straightforward. To identify specific industrial economic activities, the research employed the North American Industry Classification System (NAICS). This standardised system, developed by the USA, Mexico, and Canada, classifies workers according to the characteristics of the firm where they work. The NAICS has several codes listed in the online Appendix D. These codes help identify whether the worker is employed in a firm engaged in agricultural, industrial or service activities. Moreover, it has disaggregated codes that allow the identification of the specific economic activity of the firm in which the employee is working.

All the economic activity codes between 2110 and 2199 are used to identify mining activities. Codes 3310 and 3320 identify the manufacture of iron and steel. Code 3110 identifies the food industry. Codes 3140 and 3150 identify the manufacture of clothing and textiles. These codes are used to construct the first set of independent variables focused on the industrial sector. Therefore, these three variables are considered 'industrial subsectors'. Meanwhile, all the NAICS codes for economic activities between 6111 and 6199 identify individuals working in institutions dedicated to educational services. Codes between 6211 and 6299 identify those working in institutions dedicated to health services. Codes between 5210 and 5299 identify individuals working in institutions dedicated to financial services. These codes are used to construct the other three dependent variables, which are considered 'service subsectors'.

NAICS has clear and straightforward categories. Nevertheless, a limitation of the system is that white-collar workers in companies dedicated to industrial activities are classified as part of the industrial sector. For instance, administrative staff working in mining companies, although actually white-collar workers, are classified as workers in a mining firm.

Mexico employs another classification system, SINCO, which classifies individuals based on their job characteristics rather than their firm's characteristics. However, the SINCO classification system is not as clear as that of NAICS. For instance, these 4-digit SINCO job codes identify people working in accounting: (1212) directors and managers in accounting, financial services, banking and insurance; (1512) coordinators and area heads in accounting, financial services, banking and insurance; (2121) accountants and auditors; (2512) assistants in accounting, finance and stockbroking. Although all these individuals work in accounting, each job code falls into a different group based on their skill levels. For instance, directors in accounting are not part of an 'Accounting' group; they are part of the 'Management' group.

SINCO assigns different job codes to job groups based on job specialisation level, rather than grouping by job characteristics alone. Accordingly, for SINCO classifications, job codes are sometimes preferable to job groups. In other words, sometimes it is necessary to use specific codes to identify people with similar jobs who are not in the same group because they do not have the same level of specialisation under this classification system. Given that there are hundreds of job codes, this can be problematic. Therefore, for transparency purposes, the specific codes used to construct the independent variables are listed below. Moreover, the reader may consult the online appendix D for a review of all the SINCO codes.

To identify individuals in management jobs, all the SINCO job codes between 1111 and 1999 were used. INEGI designates this group as 'Management – Officials, Directors, and Heads'. To identify individuals working in sales, all the job codes between 4111 and 4999 were considered; INEGI labels this group as 'Sales – Traders, Sales Employees, and Sales Agents'. To identify individuals engaged in clerical work, all job codes between 3111 and 3999 were used, a group that INEGI refers to as 'Support Workers in Office Activities'. As all these categories comprise a large number of job codes, they are considered white-collar 'occupations' in this research.

In addition, my research examines women's participation in specific white-collar 'jobs'. They are considered jobs because they typically comprise only one or two job codes. Architects have job code 2263; lawyers 2135; accountants 2512 and 2121; doctors 2411 and 2412; nurses 2436, 2811 and 2821; secretaries 3111 and 3211; waiters 5116; and hairstylists 5211. These jobs were selected based on the social stigma theory and the task-content literature. For instance, Rendall (2024) considers that architecture is intensive in brain skills, waitressing relies heavily on brawn skills, and secretarial work requires high motor skills. In parallel, Acemoglu & Autor (2011) also consider some of these occupations as high in cognitive tasks or high in manual tasks. Similarly, Autor & Dorn (2013) consider some of these jobs as low-skill or high-skill services. Taken together, these complementary classifications guide the selection of occupations analysed in this study.

Third stage

The third stage of the econometric analysis evaluates another hypothesis comprising the social stigma theory: the probability that married women are working in industry or agriculture decreases as their husbands' educational level rises, although this pattern does not extend to white-collar jobs. The rationale for this hypothesis is that husbands with higher levels of education may be judged by their relatives and society if their wives work in physically demanding roles in agriculture or industry. As highly educated men typically earn higher incomes, they may prefer to bear the financial cost of their wives not working rather than a potential decline in their social status associated with their wives working in blue-collar jobs. However, this pattern does not extend to the likelihood that wives of highly educated husbands are working in white-collar jobs, as these occupations are socially accepted and do not entail physically demanding tasks.

To evaluate this hypothesis, I analyse the likelihood of wives participating in agriculture, industry, informal services and formal services, depending on their husbands' level of education. It is important to clarify that women who were single, separated, divorced, or widowed were excluded from the third stage of the analysis. The estimates pertain solely to women in stable partnerships, whether married or cohabiting. This restriction is necessary to assess whether the husband's education level is associated with their wife's probability of working in different sectors. The hypothesis is assessed using a multinomial probit regression. The structure of this third econometric model is similar to that of the two previous models, with only minor variations. The aspects that vary in this specification are outlined below, while the model itself can be expressed with the following equation:

$$P(Y_i = n) = \mathrm{F}\,(\beta_0 + \beta_g^n HusbandEducation_i + \sum_{j=4}^{j=2} \beta_j CIC_i^{(j)} + \sum_{j=8}^{j=5} \beta_j CHC_h^{(j)} + \sum_{j=10}^{j=9} \beta_j CHHC_h^{(j)} + \mu_{s,t})$$

The dependent variable 'Y' captures whether each individual *(i)* in the sample is either: 0) not working, 1) working in agriculture, 2) working in industry, 3) working in informal services, or 4) working in formal services. The category 'not working' (Y = 0) serves as the baseline for the multinomial probit estimation. Therefore, $P(Y_i = n)$ denotes the probability that a woman *(i)* is working in sector *(n)* relative to the probability of not working. In addition, *'F'* denotes the multivariate normal cumulative distribution function, while β_0 is the intercept.

The core explanatory variable of the model is '$HusbandEducation$', a categorical variable that can take one of the following values: 1) primary school or less; 2) secondary school; 3) high school; 4) technical school; or 5) university degree. In this model, primary school is used as the baseline category. In addition, the model uses sub-index 'g' to represent the level of education of the husband, so $g \in \{1, ..., 5\}$ represents the five levels of education. In this model, β_g^n will indicate the relationship between having a husband with different levels of education *(g)* and the probability of working in different sectors *(n)*. For instance, $\beta^{industry}_{university}$ will indicate whether women married to men with university degrees are more or less likely to work in industry compared with women married to men with primary school education or less (the base category). Hence, the coefficients will show whether the probability of wives working in various economic sectors decreases (or increases) if their husbands' levels of education are higher (or lower).

It should be noted that all these regressions are restricted to women in stable partnerships, whether married or cohabiting. Nevertheless, regressions are also run exclusively for men in stable partnerships for comparative purposes. In this case, the analysis focuses on whether the wife's education level is associated with a higher or lower probability of her husband working in different economic sectors.

The vector 'CIC' captures individual-level controls and includes three variables: age, age squared, and the respondent's level of education. The vector 'CHC' captures household-level controls. It comprises four variables: socioeconomic stratum, household population size, urban or rural location, and the number of children aged under five in the household. The vector 'CHHC', which accounts for household head characteristics, now includes only two variables: the household head's sex and their age. In this model, the household head's education level is excluded because it is highly correlated with the husband's education level.

The three econometric models employed in this analysis share a broadly similar structure, with only a limited number of elements differing across specifications. The purpose of this section is to guide the reader in identifying these model-specific variants. Taken together, the three econometric models provide the basis for examining the three interrelated hypotheses that constitute the social stigma theory.

Descriptive statistics

The descriptive statistics for the variables used in the regression analyses are presented in separate tables and they are included in the online Appendix A. Table 1.1 includes the dependent variables employed in the various stages of the analysis. Table 1.2 details the control variables related to individual characteristics. Table 1.3 presents the variables controlling for household characteristics, including those related to the characteristics of the household head and to the husband's education level.

This section also offers a brief overview of key insights about Mexico. It reports on the size of the industrial sector as a share of total employment, identifies the country's main manufacturing activities, and distinguishes between male-intensive and female-intensive industries. It also presents data on the marital status of working-age women. These details collectively provide context for the subsequent analysis, and they are also presented in the online Appendix.

Figure 1.11 presents the sectoral composition of jobs in Mexico. The first panel shows that, in 2019, nearly 25% of jobs in the country were in industry. The second panel shows that most of the industrial labour force works in manufacturing (67%) and construction (30%), while mining (1%) and utilities (1%) account for only a marginal share of industrial jobs.[10] Finally, the third panel disaggregates manufacturing and shows that jobs in this subsector are concentrated mainly in food processing (22.8%), auto parts and transportation equipment (14.7%), clothing (7.7%) and metal products (7.5%).

Figure 1.12 presents the gender composition of employment across industrial activities in Mexico. Construction, mining and utilities are male-intensive, while the manufacturing sector displays more gender diversity, with 62% male and 38% female workers. Disaggregating manufacturing activities enables the identification of male-dominated industries (such as metal products, timber, and machinery and equipment manufacturing), gender-balanced industries (such as food processing and the manufacture of computers and electronic components), and female-dominated industries (such as textiles and clothing).

Table 1.5 shows the distribution of the female workforce across industrial activities. Nearly 25% of women in manufacturing jobs work in food processing. Manufacturing of auto parts and transportation equipment is the second-largest employer, employing 14.2% of women who work in industrial jobs, even though these activities remain male-intensive, as almost two-thirds of the workforce are men. Meanwhile, clothing manufacturing is the third-largest employer, accounting for 12.7% of women working in industry.

Finally, Table 1.6 presents the distribution of working-age women (18–65) in Mexico by marital status: 42% are married, 28% are single, 19% are in a free union relationship. These are the three predominant marital statuses among working-age women in Mexico, collectively accounting for almost 90% of women in this age group. In contrast, 5% of working-age women are separated, 3% widowed, and 2% divorced. In the regression analyses, these three marital statuses were consolidated into a single group, named 'previously married'.

Results

As outlined in the preceding sections, the econometric analysis in this study proceeds through three distinct stages to evaluate different hypotheses related to the stigma theory on the supply of labour among women in a stable partnership. The first stage tests the hypothesis that wives, relative to single women, exhibit substantially lower probabilities of working in agriculture and

[10] *'Utilities' refers to the generation, transmission, and distribution of electricity, gas, and water.*

industry, while showing no such differential in formal services. The second stage examines two aspects of the stigma hypothesis. First, that the alleged negative relationship between married women and blue-collar jobs extends beyond male-intensive industries (e.g. construction) to gender-balanced industries (e.g. food processing) and even to female-intensive manufacturing activities (e.g. clothing and textiles). Second, that the negative relationship between marriage and working attenuates substantially—or disappears entirely—within white-collar occupations and professions that do not carry a comparable social stigma, including sales, management, finance, education and health services. The third stage of the analysis tests the hypothesis that, among wives, those married to highly educated men are less likely to work in industry and agriculture relative to women married to less educated husbands. The objective of this section is to present the empirical findings from these sequential analytical stages. To enhance readability, the main text only presents the final results, while Appendix C includes all the regression results and the details of the estimated predicted probabilities.

First stage

Goldin (1994) posits that married women exhibit sector-specific patterns of labour force participation, since there is a social stigma attached to blue-collar occupations in agriculture and industry, while white-collar service jobs remain relatively stigma-free. Following Goldin's hypothesis on stigmas and female labour supply, married women are expected to exhibit a much lower propensity to work in agriculture and industry relative to single women, whereas the difference between married and single women working in formal services should be smaller or even negligible, as the social stigma is weaker or absent in these kinds of job. My research evaluates this proposition empirically by analysing sectoral employment probabilities conditional on the marital status of Mexican women.

The first stage of the econometric analysis employs a multinomial probit regression to model the likelihood of female labour participation across five mutually exclusive employment statuses: 1) not working (baseline category), 2) working in agriculture, 3) working in industry, 4) working in informal services, or 5) working in formal services. The disaggregation of services into formal and informal categories merits further discussion. Goldin emphasised that the expansion of white-collar services facilitated the entry of married women into the labour force. Analysing the service sector as a single category would conflate formal services with informal services, thereby obscuring the mechanism hypothesised by Goldin, as informal services typically lack the professional features and social legitimacy associated with stigma-free occupations.

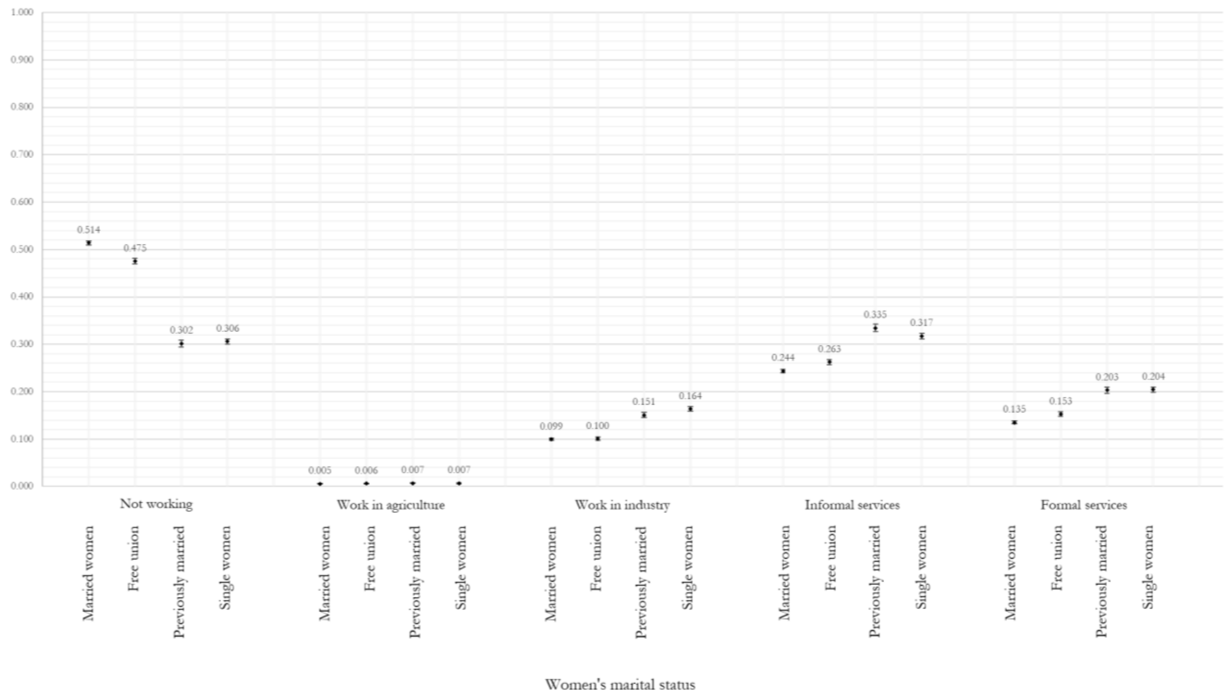
The regression results presented in Table 1.8 (see online Appendix C) reveal a pattern of negative and statistically significant coefficients for married women across all four employment categories. Hence, wives report lower probabilities of working in agriculture and industry—as anticipated—but also exhibit lower probabilities of working in both informal and formal services. Moreover, women in free-union relationships also show a similar negative relationship between having a partner and working across all sectors. These results suggest that, relative to their single counterparts, Mexican women with a permanent partner are less likely to participate in the labour market, irrespective of the economic sector.

Figure 1.1 illustrates the predicted probabilities that Mexican women are working in different economic sectors, depending on marital status. Married women exhibit consistently lower probabilities of working across all sectors relative to single women. These results challenge the stigma hypothesis, which predicts substantial differentials for wives and single women working in industry, but smaller or no differentials for wives and single women working in formal services. The results therefore do not reveal signs of a sector-specific stigma towards wives working in blue-collar jobs. They reflect a generalised pattern of a lower likelihood of working across all sectors. For comparative purposes, the predicted probabilities of Mexican men working in different economic sectors are presented in Figure 1.2, while the regression results and all the details on the estimations of predicted probabilities are presented in the online Appendix C (Tables 1.8 and 1.9).

Online appendix:

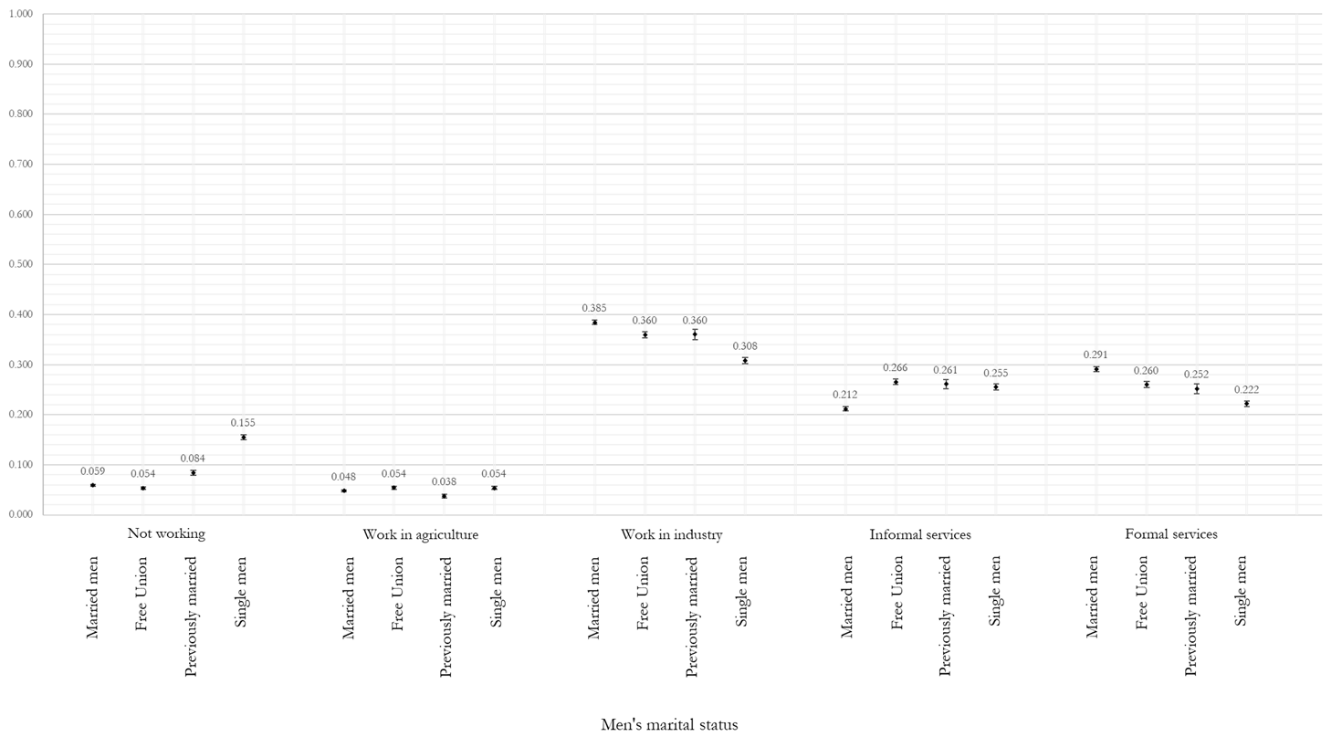
https://github.com/ilopezmoreno/onlineappendix_socialstigma_workingwives/blob/main/Online%20Appendices%20-%20Is%20there%20a%20social%20stigma%20against%20working%20wives.pdf

Figure 1.1 - Estimated probability of Mexican women working in different sectors depending on their marital status



Notes: After fitting the multinomial probit model, the predicted probabilities of working in each sector are computed for every individual in the sample, conditional on their observed individual and household characteristics, while holding all the covariates at their sample means. To prevent an overload of tables and figures, only the final results (the predicted probabilities) are presented in the main text. The regression results and the estimation of predicted probabilities are presented in Table 1.8 and Table 1.9 (see online Appendix C).

Figure 1.2 – Estimated probability of Mexican men working in different sectors depending on their marital status



Notes: After fitting the multinomial probit model, the predicted probabilities of working in each sector are computed for every individual in the sample, conditional on their observed individual and household characteristics, while holding all the covariates at their sample means. To prevent an overload of tables and figures, only the final results (the predicted probabilities) are presented in the main text. The regression results and the estimation of predicted probabilities are presented in Table 1.8 and Table 1.9 (see online Appendix C).

Second stage

Goldin (1994) posits that the alleged negative relationship between marriage and female labour force participation in industrial jobs extends across male-intensive industries (e.g. mining, iron and steel), female-intensive industries (e.g. clothing and textiles), and mixed industries (e.g. food processing). The argument implies that stigma cannot be attributed simply to the male predominance in the industrial sector, as even in industrial activities where women constitute the majority of the workforce, wives remain absent precisely because these jobs are physically demanding. In contrast, she argues that white-collar occupations within the service sector (sales, management and clerical work) are largely exempt from such social stigma, because they are regarded as socially acceptable forms of employment for married women. Goldin emphasises specific white-collar professions in education and health, explicitly mentioning teaching and nursing as acceptable pursuits for married women.

Goldin's argument rests upon the existence of a potent social stigma: the notion that only a husband who is lazy and negligent of his family would allow his wife to engage in blue-collar labour. Crucially, the central proposition is that the stigma is directed at husbands rather than at the wife herself, serving to enforce patriarchal norms of male provision, while the stigma does not extend to widows, divorced or single women performing industrial jobs.

This study examines this hypothesis and extends the empirical analysis beyond the occupations specifically discussed by Goldin to encompass a broader array of specific jobs. Rendall (2024) distinguishes between 'brawn' and 'brain' skills across different jobs, and discusses that certain jobs require physical capabilities, while others demand high cognitive competencies. She classifies waitressing as a low-skill occupation with high brawn requirements. Jobs such as firefighting and fishing are also identified as high-brawn skills in her study but are not considered because they are male-dominated activities. My research also considers jobs classified by Rendall as high in brain skills (lawyers, architects, accountants, doctors). Secretaries are also considered, whom Rendall regards as requiring high motor and brain skills, yet low brawn skills. Furthermore, the study examines nurses, mentioned by Goldin, and low-skill service jobs (hair styling) that require low cognitive skills. Therefore, this study evaluates the likelihood that married women will participate in industrial activities, white-collar occupations, and in different types of high-skill and low-skill services. The regression results are presented in Table 1.10 and Table 1.11 (see online Appendix C).

Consistent with Goldin's predictions, the results reveal a statistically significant negative relationship between marriage and female labour participation in industrial activities. This negative relationship is evident across industries such as mining (male-intensive), food processing (gender-balanced), and even clothing and textiles manufacturing (female-intensive). Nevertheless, the results also show that the negative relationship extends beyond the industrial sector. Mexican wives are also less likely to work in white-collar activities (education, health and finance), or in white-collar occupations (sales, management and clerical work). This pattern also extends to specific white-collar jobs considered in the analysis (lawyers, architects, accountants, doctors, nurses, secretaries, waitresses and hairstylists). The coefficient plot presented in Figure 1.3 illustrates this generalised negative relationship. Hence, the results suggest that married women are consistently less likely than single women to have jobs in industrial or service activities, in blue-collar or white-collar jobs, and in low-skill or high-skill services.

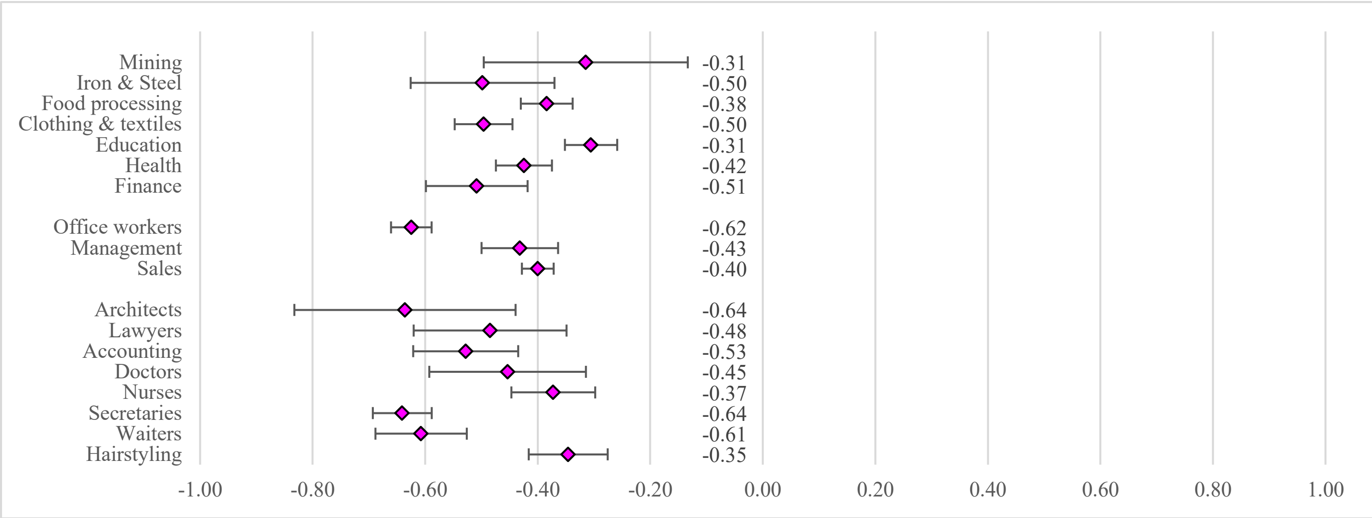
The analysis of male employment patterns yields a starkly contrasting picture. When identical regression specifications are estimated for men, the results indicate that husbands demonstrate significantly higher employment probabilities than single men across nearly all the categories examined. This positive relationship suggests that marriage serves as a catalyst for male labour force participation rather than a deterrent to it. The sole exception to this pattern emerges in hairstyling—an occupation encompassing both hairdressers and barbers. In this specific case, the results indicate that single men are more likely to work in hairstyling than husbands.

Taken together, these findings confirm that married women are less likely to work in industrial activities, but they are also less likely to hold jobs across different types of work in the service sector. The magnitude of the negative relationship is similar in industrial jobs, white-collar activities, white-collar occupations and white-collar jobs. Moreover, the results highlight a clear gender asymmetry, whereby marriage is associated with reduced labour supply among wives but increased labour supply among husbands. Hence, the results suggest once again that, rather than a sector-specific stigma attached to wives working in blue-collar jobs, married women are simply less likely to hold different types of jobs than single women.

Online appendix:

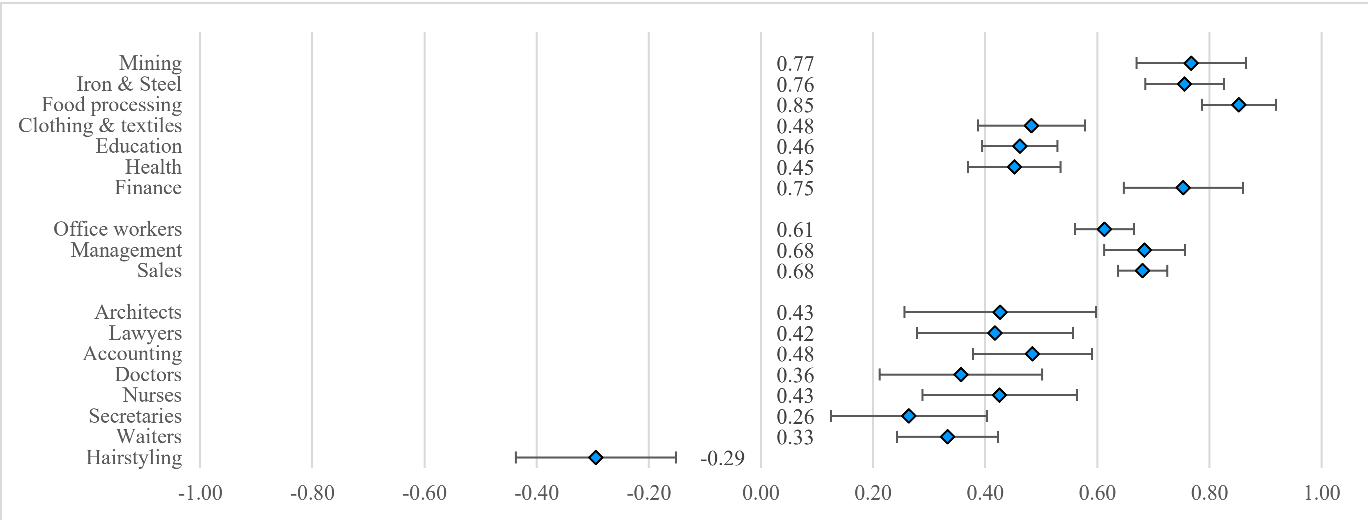
https://github.com/ilopezmoreno/onlineappendix_socialstigma_workingwives/blob/main/Online%20Appendices%20-%20Is%20there%20a%20social%20stigma%20against%20working%20wives.pdf

Figure 1.3 – Plot of regression coefficients on the likelihood that wives are working in specific industries, white-collar services, white-collar occupations, and high-skills or low-skills service jobs (compared with single women)



Notes: [*Table 1.10*]() *presents the regression results of the coefficients illustrated in this figure* [*(see online Appendix C).*]()

Figure 1.4 – Plot of regression coefficients on the likelihood that husbands are working in specific industries, white-collar services, white-collar occupations, and high-skills or low-skills service jobs (compared with single men)



Notes: [*Table 1.11*]() *presents the regression results of the coefficients illustrated in this figure* [*(see online Appendix C)*]().

Third stage

One of Goldin's social stigma hypotheses suggests that the wives of highly educated men should exhibit a considerably lower likelihood of working in agriculture or industry than the wives of less educated men. This could be for a variety of reasons: first, wives' employment in these sectors is stigmatised because of its physically demanding nature; second, such work could be considered a signal that the wife originates from a lower socioeconomic stratum; third, highly educated men might not permit their wives to work in these sectors, as doing so diminishes the family's social status. Conversely, highly educated men would not view their wives' employment in white-collar services unfavourably for different reasons: first, such work is not physically demanding; second, society would not judge the husband as negligent; third, women value having a formal job that is not physically demanding and aligns with their level of education.

This section presents the empirical results derived from testing this hypothesis in the Mexican labour market. It is important to note that women who are single, separated, divorced or widowed were excluded from the analysis. The estimates focus on women in permanent partnerships, whether married or in free union relationships. Nonetheless, for simplicity in interpreting the results, the terms 'wives' and 'husbands' are used throughout the discussion.

The results of the regression analysis are presented in [Table 1.12 (see online Appendix C).]() The first four columns of the table report the estimated coefficients associated with the probability that a married woman is working in (1) agriculture, (2) industry, (3) informal services, or (4) formal services, conditional on her husband's level of education. It should be noted that these coefficients are derived from a multinomial probit regression in which the base outcome is (0) 'not working'. In addition, Figure

1.5 illustrates the estimated probability that married women do not work, or that they work in any of these four main sectors. For comparative purposes, Figure 1.6 shows the predicted probabilities for husbands. These estimates were derived from the multinomial regression model fitted in the third stage of the econometric analysis.

The predicted probabilities of wives, shown in Figure 1.5, reveal interesting patterns. First, the higher the husband's educational level, the lower the likelihood that married women are working. Similarly, the probability of wives working in agriculture, industry or informal services decreases monotonically as husbands' education increases. In contrast, the probability of married women working in formal services is the only one to show the opposite trend, as it does not decrease with the husband's education; instead, it increases slightly among women married to highly educated men.

Of all the empirical evaluations conducted in this research, this is the only aspect of Goldin's social stigma theory that holds in the Mexican labour market. Nonetheless, these results must be interpreted with caution. The household profile that aligns with the predicted pattern—highly educated husbands and wives employed in formal services—constitutes only a marginal proportion of the Mexican population. Far more prominent is the tendency for wives not to participate in the labour market, which reaches its highest level in households where the husband holds a university degree. More precisely, in households where the husband has a university degree, the probability of wives not participating in the labour market is around 62%, while their probability of working in formal services is only around 12%.

If we examine these results through the lens of social stigma theory, one could argue that the marginally higher propensity of wives married to highly educated men to work in formal services may be reflecting husbands' preferences for white-collar jobs over blue-collar employment for their wives. Nevertheless, given the descriptive nature of this research, we cannot conclusively attribute this pattern to social stigma. An equally plausible explanation is assortative mating: women married to highly educated men typically possess comparable educational credentials and are therefore more inclined to seek formal service-sector positions that align with their qualifications and career aspirations, rather than taking jobs in agriculture or industry. Regardless of the underlying mechanism, it is essential to recognise that this result pertains to a highly delimited subset of households. Instead, attention should be directed towards the dominant pattern in the analysis: households with highly educated husbands exhibit the greatest likelihood of wives remaining outside the labour market.

The observed pattern of low labour market participation among wives married to highly educated men also warrants careful interpretation, as it may arise from multiple underlying mechanisms. These findings do not necessarily imply that highly educated husbands actively discourage or prevent their wives from working. In some households, standard income effects may permit single-earner arrangements. In others, shared preferences for a 'traditional' division of labour—husbands working and wives remaining at home—may prevail, even when household income is not particularly high. In yet other cases, highly educated husbands may stigmatise their wives' engagement in paid work, or the wives themselves may stigmatise having to work. The dynamics of the marriage market also merit consideration: some women may choose to marry highly educated men with the intention of withdrawing from the labour market, while in other cases, highly educated men may expect or request their wives to exit employment upon marriage.

It lies beyond the scope of this study to infer personal motivations or intra-household bargaining processes. The aim of the analysis is, instead, to assess the empirical relevance of Goldin's hypothesis that a strong social stigma discourages wives from working in blue-collar jobs. This is particularly relevant due to the considerable emphasis Goldin places on this stigma theory to explain the low LFPRs among married women in middle-income countries like Mexico and also because there are no prior studies that provide an empirical assessment of the theory.

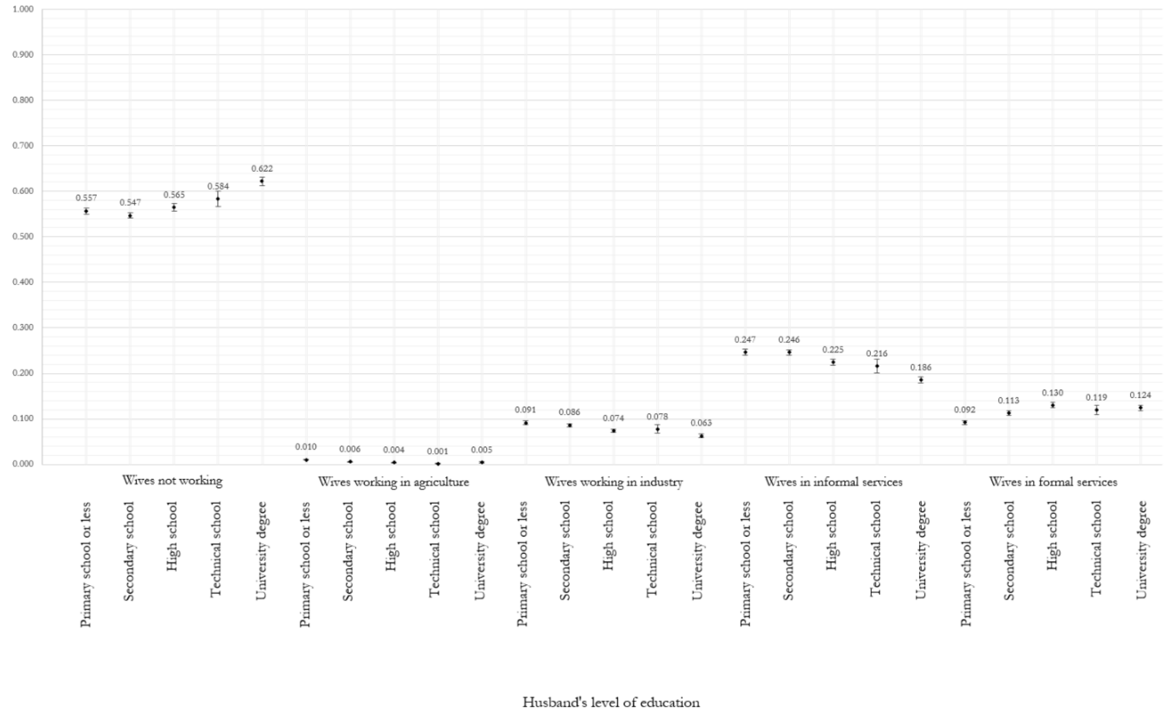
Table 1.12 (in online Appendix C) also includes the results of the variable controlling for the household socioeconomic stratum. This variable is included in the model as a proxy of the income effect hypothesised by Goldin, whereby higher household income reduces a wife's labour supply. However, 'stratum' is a categorical proxy based primarily on household assets. Hence, it cannot be used to detect marginal changes in husbands' earnings that might generate meaningful income effects on wives' labour supply. To evaluate Goldin's income-effect hypothesis, a different type of analysis is required—one that employs panel data and examines whether an increase in the husband's income induces the withdrawal of wives from the labour market. This lies beyond the scope of this study. The central focus of the third stage of the analysis is to evaluate the hypothesis that wives of highly educated husbands are less likely to work in agriculture and industry, but that this negative relationship does not extend to white-collar services. Consequently, the analysis focused on the husband's level of education to assess this hypothesis.

Overall, the findings do not show signs of a strong social stigma against wives working in blue-collar jobs. Rather, they suggest the existence of a broader set of constraints, preferences, and social norms that shape the labour-market participation of married women across all sectors, with the partial exception of formal services, where married women with highly educated husbands display a slightly higher likelihood of participation compared with those married to men with lower levels of education.

Online appendix:

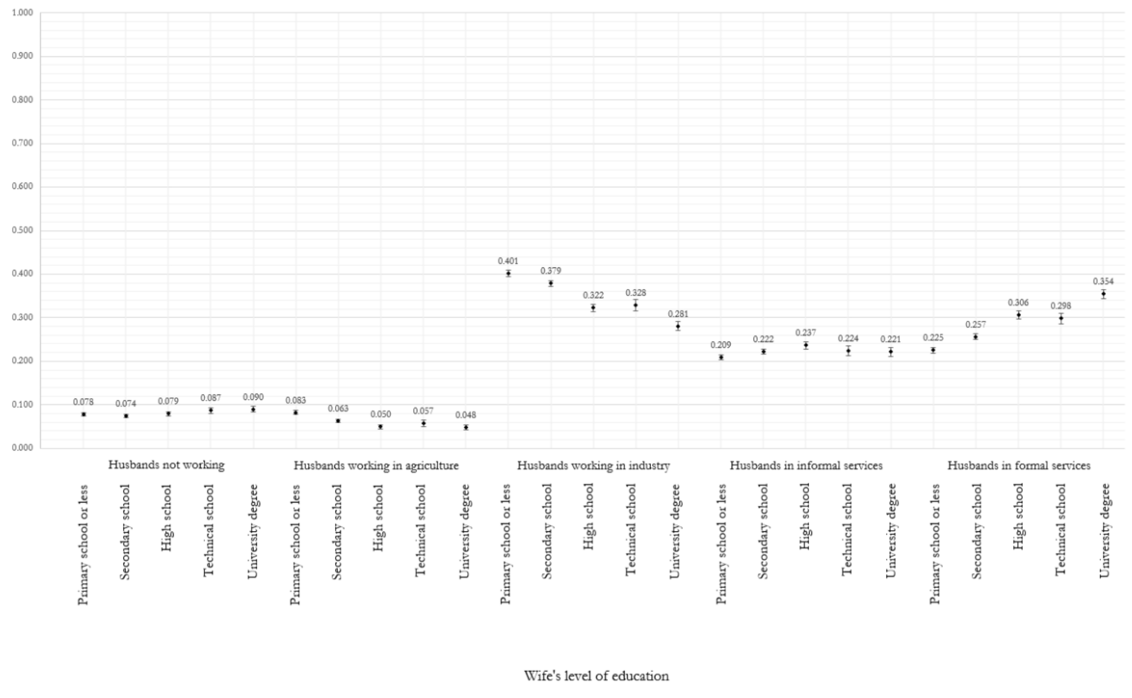
https://github.com/ilopezmoreno/onlineappendix_socialstigma_workingwives/blob/main/Online%20Appendices%20-%20Is%20there%20a%20social%20stigma%20against%20working%20wives.pdf

Figure 1.5 – Predicted probabilities of wives working in different economic sectors depending on their husbands' education



Notes: After fitting the multinomial probit model, the predicted probabilities of working in each sector are computed for every individual in the sample, conditional on their observed individual and household characteristics, while holding all the covariates at their sample means. To prevent an overload of tables and figures, only the final results (the predicted probabilities) are presented in the main text. The regression results and the estimations of predicted probabilities are presented in Tables 1.12 and 1.13, respectively (see online Appendix C)

Figure 1.6 – Predicted probabilities of husbands working in different economic sectors depending on their wives' education



Notes: After fitting the multinomial probit model, the predicted probabilities of working in each sector are computed for every individual in the sample, conditional on their observed individual and household characteristics, while holding all the covariates at their sample means. To prevent an overload of tables and figures, only the final results (the predicted probabilities) are presented in the main text. The regression results and the estimations of predicted probabilities are presented in Tables 1.12 and 1.14, respectively (see online Appendix C).

Discussion

Readers of this research may find the previous results compelling, yet insufficient to consider marriage as a relevant determinant of female labour force non-participation. Fortunately, the ENOE household survey includes a direct question that allows respondents to indicate their main reason for leaving their last job. Table 1.7 documents the most selected responses, differentiating by gender (see online Appendix B). Among men, the most popular reason is the pursuit of higher earnings, selected by 33% of male respondents. In contrast, 60% of women attributed their withdrawal from their last job to “marriage, pregnancy, or family responsibilities”, making this the overwhelmingly dominant response among female respondents.

It is essential to recognise that the most frequently selected response among women—“marriage, pregnancy, or family responsibilities”—is a composite category. This aggregation precludes a precise assessment of whether it is marriage, pregnancy or domestic responsibilities that is the principal driver of women’s labour-market withdrawal. Greater analytical clarity would be achieved if INEGI disaggregated these options in future survey rounds, thereby enabling a more accurate identification of the underlying cause. Notwithstanding this limitation, the data remain informative: they strongly suggest that marriage plays a role in women’s labour-market exit in Mexico, even if it is not possible to ascertain whether it is the main factor. Nevertheless, cross-country evidence from Kleven et al. (2025) offers relevant context, as these authors examine whether motherhood or marriage is the primary determinant of lower female labour force participation in different countries. They find that, in terms of female labour participation, the marriage penalty is stronger than the child penalty in most developing countries—including Mexico—while the child penalty is stronger than the marriage penalty in most high-income countries.

Irrespective of whether marriage, pregnancy or family responsibilities were the main reason for women to leave their last job, a separate finding emerges from the data on the reasons for labour market inactivity among Mexican women. Official statistics indicate that the principal reason preventing non-working women from participating in the labour market is the absence of care-giving support for children, the elderly, ill or disabled individuals living in the household (see Figure 1.13 in online Appendix B). The ENOE survey asks all women this question, even those who have never worked. Therefore, although the data cannot determine whether marriage, pregnancy, or family responsibilities are the main reasons for female labour market withdrawal, the evidence indicates that caregiving responsibilities are the primary constraint for Mexican women to supply labour.

In light of the preceding discussion, readers must consider that this analysis has some limitations. Therefore, they should refrain from drawing causal claims from it. A key constraint of the analysis is the potential endogeneity between female labour participation and marital status, as these variables are subject to simultaneous causality. This section explains why the study does not address this limitation but rather explicitly acknowledges it as a boundary of the analysis.

The study draws on a dataset of more than 500,000 women surveyed between 2016 and 2019. Undoubtedly, there are women in the dataset who have chosen to divorce their husbands because they were pressured to quit their jobs and dedicate themselves entirely to household duties. There will also be cases of women who are currently single because their priority is to work, and who are not interested in having a romantic relationship. There will also be cases of women living in very conservative environments who have chosen to remain single because, if they were to marry, their spouse would probably ask them to stop working. All these examples illustrate women prioritising their careers over being married. Hence, in these cases, the current labour statuses of these women act as a causal factor shaping their current marital status.

Conversely, some women in the dataset may have chosen to prioritise marriage over having a job. Some may have put their careers on hold because of a strong desire to marry and start a family, while other women may have married men who have the financial capacity to withdraw their wives from the labour market just after marriage. In these settings, marital choices have induced a specific labour status—namely, non-participation in the workforce. Hence, marital status is a causal factor that shapes their current employment status in these cases.

In other situations, some women married a man who, over time, encouraged them to stop working— either because of the birth of a child or as a result of jealousy. Some women thus married without knowing that their husbands would ask them to leave their jobs. In some of these cases, women may choose to remain married and not work rather than divorce; in others, they may prefer to divorce and continue working. Hence, in some of these cases, their marital status was influenced by their current employment status, while in other cases, their desired employment status shaped their current marital status.

These examples demonstrate that the dataset contains numerous complex and nuanced stories shaping women’s decisions about marriage and employment. Attempting to disentangle the factors influencing these decisions would be an overly ambitious and speculative endeavour. It is beyond the scope of this study to infer intra-household discussions or women’s personal thoughts. This would probably require either fieldwork research or, alternatively, the development of questions designed by INEGI to pinpoint the main reason married women do not participate in the labour market.

Notwithstanding these limitations, the descriptive evidence presented in this study yields an important substantive conclusion: the lack of labour-market participation among married women in Mexico cannot be primarily attributed to a sector-specific stigma. The central aim of this investigation is to assess whether the Mexican labour market shows signs of a strong stigma against wives working in blue-collar jobs. More broadly, the objective is also to provide the first empirical evaluation of the social-stigma theory. After conducting this research, the findings point to a different pattern. The evidence indicates that Mexican wives are highly unlikely to have a job. More specifically, the results indicate that, compared with single women, Mexican wives are less likely to hold either a blue-collar job or a white-collar job.

Conclusion

This chapter provides a comprehensive empirical evaluation of Goldin's social stigma hypothesis, which posits that low female labour force participation rates (FLPRs) in middle-income economies, such as Mexico, may be driven by a sector-specific stigma against wives working in physically demanding blue-collar jobs. Hence, the main goal is to examine whether Mexico's labour market reflects a strong social stigma towards wives working in the industrial sector. One of the study's most significant findings is the discovery of a strong negative association between having a partner and having a job across all economic sectors—agriculture, industry, formal services, and informal services alike. The study also presents a granular analysis showing that this negative relationship is homogeneous not only across sectors but also across different types of industries, white-collar occupations, and several high-skill professions and low-skill service jobs.

These results challenge the central premise of the stigma hypothesis, which posits a concentrated resistance to wives' employment in blue-collar occupations. Therefore, this finding contributes to the literature by contesting the assumption that the negative association towards wives working in industrial jobs attenuates or disappears in white-collar services, which are conventionally viewed as stigma-free jobs for married women. The results thus indicate that barriers to married women's labour market participation in Mexico transcend sectoral boundaries and extend well beyond physically demanding work and blue-collar occupations. Rather, the evidence reveals a generalised pattern in which married women exhibit homogeneous lower probabilities of employment across different types of jobs.

These findings suggest an enduring prevalence of traditional household divisions of labour in married-couple households in Mexico, where husbands engage in paid employment while wives typically remain outside the labour market. Survey evidence from Mexico also shows that "marriage, pregnancy, and family responsibilities" are the predominant reasons why women left their last job. The findings thus align with research indicating that domestic responsibilities, childcare demands, and marriage norms are key determinants of wives' labour supply patterns.

On the other hand, other mechanisms could account for these patterns. Within each Mexican household, various factors—operating either simultaneously or in isolation—may explain married women's non-participation in the labour market. These include the prevalence of breadwinner norms within the household, strong income effects arising from husbands' earnings, motherhood penalties, and patterns of assortative mating in the marriage market. Importantly, these mechanisms do not necessarily operate independently; households may exhibit overlapping dynamics where traditional gender norms coincide with economic incentives or caregiving demands.

Disentangling the relative contribution of each factor to observed employment patterns remains an important avenue for future research. Additionally, comparative studies are also needed to determine whether Mexico is an outlier in exhibiting a generalised negative relationship between marriage and female employment across sectors, or if this pattern holds true in other middle-income countries, thereby necessitating a fundamental revision of the stigma theory.

The policy implications of these findings merit consideration. Given that the negative association between marriage and employment persists across the labour market—spanning both formal and informal sectors, blue-collar and white-collar occupations, and low-skill and high-skill positions—interventions should address the gender norms and traditional household roles that constrain married women's participation. The evidence suggests that policies aimed at relaxing household constraints and enhancing the compatibility between market work and domestic responsibilities—such as expanded childcare provision—could shift the labour supply patterns of married women.

To conclude, this research highlights its main contributions to the literature. First, it provides the first comprehensive empirical evaluation of the social stigma hypothesis against working wives. Second, it extends beyond Goldin's original framework by examining not only formal service-sector employment but also informal services—an important distinction in developing countries—while demonstrating that the negative association between marriage and female employment persists in both types of service jobs. Third, by studying wives' likelihood of participation across industries, activities, occupations, and task-content categories—including jobs intensive in cognitive, manual, and motor skills—the study offers a granular and detailed empirical analysis of the stigma theory. Finally, the findings challenge the premise that a strong stigma towards blue-collar jobs is behind the lower FLPRs displayed by wives relative to single women. Instead, the evidence reveals a generalised pattern whereby married women in Mexico exhibit homogenous lower probabilities of employment across diverse job types, suggesting that barriers to their labour market participation are systemic rather than sector-specific.

Online appendix:

https://github.com/ilopezmoreno/onlineappendix_socialstigma_workingwives/blob/main/Online%20Appendices%20-%20Is%20there%20a%20social%20stigma%20against%20working%20wives.pdf

References

Aaronson, D., Dehejia, R., Jordan, A., Pop-Eleches, C., Samii, C., & Schulze, K. (2021). The Effect of Fertility on Mothers' Labor Supply over the Last Two Centuries. *The Economic Journal*, *131*(633), 1–32. https://doi.org/10.1093/ej/ueaa100

Acemoglu, D., & Autor, D. (2011). Skills, Tasks and Technologies: Implications for Employment and Earnings. *Handbook of Labor Economics*, *4B*, 1043–1171.

Aguayo-Tellez, E., Airola, J., & Juhn, C. (2010). *Did Trade Liberalization Help Women? The Case of Mexico in the 1990s* (Working Paper No. 16195). National Bureau of Economic Research. https://doi.org/10.3386/w16195

Arceo-Gomez, E., & Campos-Vazquez, R. (2010). *Labor supply of married women in Mexico: 1990-2000* [Serie documentos de trabajo del Centro de Estudios Económicos]. El Colegio de México, Centro de Estudios Económicos. https://econpapers.repec.org/paper/emxceedoc/2010-16.htm

Atkin, D. (2016). Endogenous Skill Acquisition and Export Manufacturing in Mexico. *American Economic Review*, *106*(8), 2046–2085. https://doi.org/10.1257/aer.20120901

Autor, D. H., & Dorn, D. (2013). The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market. *American Economic Review*, *103*(5), 1553–1597. https://doi.org/10.1257/aer.103.5.1553

Becker, A. (2025). On the Economic Origins of Concerns Over Women's Chastity. *The Review of Economic Studies*, *92*(4), 2303–2329. https://doi.org/10.1093/restud/rdae084

Becker, G. S. (1973). A Theory of Marriage: Part I. *Journal of Political Economy*, *81*(4), 813–846.

Becker, G. S. (1993). *A Treatise on the Family: Enlarged Edition*. Harvard University Press.

Berniell, I., Berniell, L., Mata, D. de la, Edo, M., & Marchionni, M. (2021). Gender gaps in labor informality: The motherhood effect. *Journal of Development Economics*, *150*, 102599. https://doi.org/10.1016/j.jdeveco.2020.102599

Bhalotra, S. R., & Fernández, M. (2021). *The rise in women's labour force participation in Mexico: Supply vs demand factors* (Working Paper No. 2021/16). WIDER Working Paper. https://doi.org/10.35188/UNU-WIDER/2021/950-1

Blank, R. M., & Shierholz, H. (2006). *Exploring Gender Differences in Employment and Wage Trends Among Less-Skilled Workers* (Working Paper No. 12494). National Bureau of Economic Research. https://doi.org/10.3386/w12494

Blau, F. D., & Kahn, L. M. (2007). Changes in the Labor Supply Behavior of Married Women: 1980–2000. *Journal of Labor Economics*, *25*(3), 393–438. https://doi.org/10.1086/513416

Blau, F. D., & Kahn, L. M. (2017). The Gender Wage Gap: Extent, Trends, and Explanations. *Journal of Economic Literature*, *55*(3), 789–865. https://doi.org/10.1257/jel.20160995

Boserup, E. (1970). *Woman's Role in Economic development.* London: George Allen and Unwin Ltd.

Burda, M., Hamermesh, D. S., & Weil, P. (2013). Total work and gender: Facts and possible explanations. *Journal of Population Economics*, *26*(1), 239–261. https://doi.org/10.1007/s00148-012-0408-x

Campos-Vazquez, R. M., & Velez-Grajales, R. (2014). Female Labour Supply and Intergenerational Preference Formation: Evidence for Mexico. *Oxford Development Studies*, *42*(4), 553–569. https://doi.org/10.1080/13600818.2014.900006

Caraway, T. L. (2007). *Assembling Women: The Feminization of Global Manufacturing*. Cornell University Press.

Chant, S. H. (1991). *Women and Survival in Mexican Cities: Perspectives on Gender, Labour Markets, and Low-income Households*. Manchester University Press.

Cooray, A., Gaddis, I., & Wacker, K. M. (2012). *Globalization and Female Labor Force Participation in Developing Countries: An Empirical (Re-)Assessment* (Working Paper No. 129). Discussion Papers. https://www.econstor.eu/handle/10419/90454

Cunningham, W. V. (2001). Breadwinner or caregiver? How household role affects labor choices in Mexico. *Policy Research Working Paper Series, Policy Research Working Paper Series*, Article 2743. https://ideas.repec.org//p/wbk/wbrwps/2743.html

Datta Gupta, N., Nandy, D., & Siddhanta, S. (2020). "Opt out" or kept out? The effect of stigma, structure, selection, and sector on the labor force participation of married women in India. *Review of Development Economics*, *24*(3), 927–948. https://doi.org/10.1111/rode.12672

Ederington, J., Minier, J., & Troske, K. R. (2009). *Where the girls are: Trade and labor market segregation in Colombia* (Working Paper No. 4131). IZA Discussion Papers. https://www.econstor.eu/handle/10419/35425

England, P., Garcia-Beaulieu, C., & Ross, M. (2004). Women's Employment among Blacks, Whites, and Three Groups of Latinas: Do More Privileged Women Have Higher Employment? *Gender & Society*, *18*(4), 494–509. https://doi.org/10.1177/0891243204265632

Erten, B., & Keskin, P. (2024). Trade-offs? The Impact of WTO Accession on Intimate Partner Violence in Cambodia. *The Review of Economics and Statistics*, *106*(2), 322–333. https://doi.org/10.1162/rest_a_01140

Fatima, A., & Sultana, H. (2009). Tracing out the U-shape relationship between female labor force participation rate and economic development for Pakistan. *International Journal of Social Economics*, *36*(1–2), 182–198. https://doi.org/10.1108/03068290910921253

Fernández, R. (2013). Cultural Change as Learning: The Evolution of Female Labor Force Participation over a Century. *American Economic Review*, *103*(1), 472–500. https://doi.org/10.1257/aer.103.1.472

Fernández, R., Fogli, A., & Olivetti, C. (2004). Mothers and Sons: Preference Formation and Female Labor Force Dynamics. *The Quarterly Journal of Economics*, *119*(4), 1249–1299. https://doi.org/10.1162/0033553042476224

Fernández-Kelly, M. P. (1983). *For We are Sold, I and My People: Women and Industry in Mexico's Frontier*. SUNY Press.

Fogli, A., & Veldkamp, L. (2011). Nature or Nurture? Learning and the Geography of Female Labor Force Participation. *Econometrica*, *79*(4), 1103–1138. https://doi.org/10.3982/ECTA7767

Fontana, M. (2003). The gender effects of trade liberalization in developing countries: A review of the literature. *Discussion Papers in Economics*, *101*, 1–29.

Gaddis, I., & Klasen, S. (2013). Economic development, structural change, and women's labor force participation: A Reexamination of the Feminization U Hypothesis. *Journal of Population Economics*, *27*(3), 639–681. https://doi.org/10.1007/s00148-013-0488-2

Gaddis, I., & Pieters, J. (2017). The Gendered Labor Market Impacts of Trade Liberalization: Evidence from Brazil. *Journal of Human Resources*, *52*(2), 457–490. https://doi.org/10.3368/jhr.52.2.1014-6690R1

Goldin, C. (1994). The U-Shaped Female Labor Force Function in Economic Development and Economic History. *NBER Working Paper Series, Working Paper Series*, (Working Paper No. 4707). https://doi.org/10.3386/w4707

Goldin, C. (2006). The Quiet Revolution That Transformed Women's Employment, Education, and Family. *American Economic Review*, *96*(2), 1–21. https://doi.org/10.1257/000282806777212350

Heath, R., & Jayachandran, S. (2016). *The Causes and Consequences of Increased Female Education and Labor Force Participation in Developing Countries* (Working Paper No. 22766). National Bureau of Economic Research. https://doi.org/10.3386/w22766

Hein, C. R. (1986). The Feminisation of Industrial Employment in Mauritius: A Case of Sex Segregation. In *Sex Inequalities in Urban Employment in the Third World* (pp. 277–311). https://www.africabib.org/rec.php?RID=W00077325

INEGI. (2014). *La informalidad laboral. Encuesta Nacional de Ocupación y Empleo. Marco conceptual y metodológico*. ISBN: 978-607-739-109-8. https://www.inegi.org.mx/app/biblioteca/ficha.html?upc=702825060459

Jayachandran, S. (2020). *Social Norms as a Barrier to Women's Employment in Developing Countries* (Working Paper No. 27449). National Bureau of Economic Research. https://doi.org/10.3386/w27449

Jensen, R. (2012). Do Labor Market Opportunities Affect Young Women's Work and Family Decisions? Experimental Evidence from India *. *The Quarterly Journal of Economics*, *127*(2), 753–792. https://doi.org/10.1093/qje/qjs002

Juhn, C., Ujhelyi, G., & Villegas-Sanchez, C. (2014). Men, women, and machines: How trade impacts gender inequality. *Journal of Development Economics*, *106*, 179–193. https://doi.org/10.1016/j.jdeveco.2013.09.009

Jütting, J., Luci, A., & Morrisson, C. (2010). Why Do so many Women End up in Bad Jobs?: A Cross-country Assessment. *OECD Development Centre Working Papers*, *2010*. https://doi.org/10.1787/5kmlhlrz6br0-en

Klasen, S. (2019). What Explains Uneven Female Labor Force Participation Levels and Trends in Developing Countries? *The World Bank Research Observer*, *34*(2), 161–197. https://doi.org/10.1093/wbro/lkz005

Kleven, H., Landais, C., & Leite-Mariante, G. (2025). The Child Penalty Atlas. *The Review of Economic Studies*, *92*(5), 3174–3207. https://doi.org/10.1093/restud/rdae104

Kucera, D., & Tejani, S. (2014). Feminization, Defeminization, and Structural Change in Manufacturing. *World Development*, *64*, 569–582. https://doi.org/10.1016/j.worlddev.2014.06.033

Majlesi, K. (2016). Labor market opportunities and women's decision making power within households. *Journal of Development Economics*, *119*, 34–47. https://doi.org/10.1016/j.jdeveco.2015.10.002

Maloney, W. F. (2004). Informality Revisited. *World Development*, *32*(7), 1159–1178. https://doi.org/10.1016/j.worlddev.2004.01.008

Mammen, K., & Paxson, C. (2000). Women's Work and Economic Development. *Journal of Economic Perspectives*, *14*(4), 141–164. https://doi.org/10.1257/jep.14.4.141

Mansour, H., Medina, P., & Velásquez, A. (2022). Import competition and gender differences in labor reallocation. *Labour Economics*, *76*, 102149. https://doi.org/10.1016/j.labeco.2022.102149

Melitz, M. J. (2003). The Impact of Trade on Intra-Industry Reallocations and Aggregate Industry Productivity. *Econometrica*, *71*(6), 1695–1725. https://doi.org/10.1111/1468-0262.00467

Mincer, J. (1962). Labor Force Participation of Married Women: A Study of Labor Supply. In *Aspects of Labor Economics* (pp. 63–105). Princeton University Press. https://www.nber.org/books-and-chapters/aspects-labor-economics/labor-force-participation-married-women-study-labor-supply

Muñoz Boudet, A. M., Petesch, P., Turk, C., & Thumala, A. (2013). On Norms and Agency: Conversations about Gender Equality with Women and Men in 20 Countries. *World Bank Publications - Books*. https://ideas.repec.org//b/wbk/wbpubs/13818.html

Munshi, K., & Singh, S. (2024). *Social Status, Economic Development and Female Labor Force (Non) Participation* (Working Paper No. 32946). National Bureau of Economic Research. https://doi.org/10.3386/w32946

Nayyar, G., Hallward-Driemeier, M., & Davies, E. (2021). *At Your Service?: The Promise of Services-Led Development*. World Bank Publications.

Ngai, L. R., & Petrongolo, B. (2017). Gender Gaps and the Rise of the Service Economy. *American Economic Journal: Macroeconomics*, *9*(4), 1–44. https://doi.org/10.1257/mac.20150253

Olivetti, C. (2013). *The Female Labor Force and Long-run Development: The American Experience in Comparative Perspective* (Working Paper No. 19131). National Bureau of Economic Research. https://doi.org/10.3386/w19131

Ozler, S. (2000). Export Orientation and Female Share of Employment: Evidence from Turkey. *World Development*, *28*(7), 1239–1248. https://doi.org/10.1016/S0305-750X(00)00034-6

Papanek, H. (1979). Family Status Production: The 'Work' and 'Non-Work' of Women. *Signs: Journal of Women in Culture and Society*, *4*(4), 775–781. https://doi.org/10.1086/493663

Perry, G. E., Maloney, W., Arias, O., Fajnzylber, P., Mason, A. D., & Saavedra, J. (2007). *Informality: Exit and Exclusion*. The World Bank Group. https://EconPapers.repec.org/RePEc:wbk:wbpubs:6730

Rendall, M. (2024). Brain versus Brawn: The Realisation of Women's Comparative Advantage. *CEPR Discussion Papers, CEPR Discussion Papers*, Article 18825. https://ideas.repec.org//p/cpr/ceprdp/18825.html

Roncolato, L. (2016). The Feminization U in South Africa: Economic Structure and Women's Labor Force Participation. *Feminist Economics*, *22*(4), 54–81. https://doi.org/10.1080/13545701.2016.1172721

Seguino, S., & Grown, C. (2006). Gender equity and globalization: Macroeconomic policy for developing countries. *Journal of International Development*, *18*(8), 1081–1104. https://doi.org/10.1002/jid.1295

Standing, G. (1999). Global Feminization Through Flexible Labor: A Theme Revisited. *World Development*, *27*(3), 583–602. https://doi.org/10.1016/S0305-750X(98)00151-X

Standing, H. (1991). *Dependence and Autonomy: Women's Employment and the Family in Calcutta*. Routledge.

Suleiman, H. (2025). The Feminization U in Egypt: Economic Development, Structural Change, and Female Labour Force Decline. *The Journal of Development Studies*, *0*(0), 1–28. https://doi.org/10.1080/00220388.2025.2563854

Tam, H. (2011). U-shaped female labor participation with economic development: Some panel data evidence. *Economics Letters*, *110*(2), 140–142. https://doi.org/10.1016/j.econlet.2010.11.003

Tejani, S., & Milberg, W. (2016). Global Defeminization? Industrial Upgrading and Manufacturing Employment in Developing Countries. *Feminist Economics*, *22*(2), 24–54. https://doi.org/10.1080/13545701.2015.1120880

Uberti, L. J., & Douarin, E. (2023). The Feminisation U, cultural norms, and the plough. *Journal of Population Economics*, *36*(1), 5–35. https://doi.org/10.1007/s00148-022-00890-5

Weinberg, B. A. (2000). Computer Use and the Demand for Female Workers. *ILR Review*, *53*(2), 290–308. https://doi.org/10.1177/001979390005300206

Weiss, Y., & Fershtman, C. (1998). Social status and economic performance: A survey. *European Economic Review*, *42*(3), 801–820. https://doi.org/10.1016/S0014-2921(97)00137-2

Welter, B. (1966). The Cult of True Womanhood: 1820-1860. *American Quarterly*, *18*(2), 151–174. https://doi.org/10.2307/2711179

Yellen, J. L. (2020). The history of women's work and wages and how it has created success for us all. *Brookings*. https://www.brookings.edu/articles/the-history-of-womens-work-and-wages-and-how-it-has-created-success-for-us-all/

Yu, Z., Wu, X., Li, M., & Guo, R. (2021). Import competition and the gender gap in labor force participation: Evidence from China. *China Economic Review*, *69*, 101689. https://doi.org/10.1016/j.chieco.2021.101689